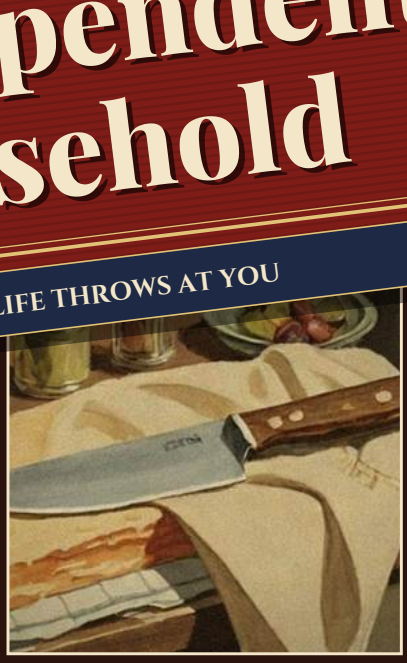







The Independent Household

BE READY FOR ANYTHING LIFE THROWS AT YOU






FORGOTTEN 60 SKILLS

THE INDEPENDENT HOUSEHOLD



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INDEPENDENT HOUSEHOLD

60 Forgotten Skills for Self-Reliant Living

AN ILLUSTRATED HANDBOOK OF OLD-TIME KNOW-HOW

HOME EDITION · 2026

The Independent Household: 60 Forgotten Skills for Self-Reliant Living. Home Edition, 2026.

Written and illustrated as an original work. The text and drawings in this book were created from scratch for this edition and are not copied from any other publication.

Please read before using this book. The skills here are offered for general education and are drawn from long-standing traditional practice and public guidance from food-safety, public-health and emergency agencies. They are not a substitute for training, local laws, or the advice of a doctor, licensed electrician, or other professional. Food preservation, fire, tools, chemicals and first aid all carry real risks. Follow current tested guidance (for example, USDA canning guides), check the laws where you live, supervise children, and when in doubt, get expert help. Home remedies in this book are traditional comfort measures, not medical treatment. In an emergency, call 911.

Printing tips. This book is laid out on US Letter (8.5 x 11 in) pages. Print double-sided, “flip on long edge”, at 100% scale. For a book that lies flat like a reference manual, take it to a copy shop and ask for spiral or comb binding with clear front and heavy back covers.

Made with care for people who still like to do things themselves.

For the grandparents who knew how,
and the grandchildren who want to learn.

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INTRODUCTION

How to use this book

There was a time, not so long ago, when the word *independence* described a household as much as a nation. A family with a well, a garden, a root cellar, a woodpile and a few hens could ride out a hard winter, a lost job or a washed-out road. The skills that made that possible were not secret. They were passed hand to hand, from parent to child, at the kitchen table and in the barn.

In two generations most of that know-how has slipped away. We have gained a great deal: safe city water, refrigerators, antibiotics, electric light at the flick of a switch. But we've also become dependent on long chains of trucks, power lines and stores that can break during a storm, a blackout or simply a bad month. This book is a small step back toward resilience.

What you'll find inside. Sixty skills in nine chapters, each on a single page laid out the same way: a picture for every step, a short explanation of why it matters, a materials panel with what you'll need, and a notes panel with tips and the mistakes to avoid. The tab on the edge of each page tells you which chapter and skill you are looking at, so you can find your way quickly when the lights are out and you're reading by lantern.

How to use it. Don't wait for an emergency to try a skill for the first time. Pick one each week. Bake the soda bread. Fill the water jugs. Sharpen the kitchen knives. Practice the car escape with your kids in the driveway. Each skill you actually do adds to a quiet confidence that's worth more than any gadget.

Start with the basics. If you're new to all this, begin with chapter 1 (water), the home medical kit in chapter 4, the winter blackout plan in chapter 5 and the family emergency plan in chapter 8. Those four alone will put your household ahead of most.

The appendix at the back gathers the tables and checklists you'll reach for most: how much bleach to add to water, how long pantry foods keep, packing lists for a go-bag and a two-week stay at home, common wild edibles, kitchen conversions, a glossary, and a fill-in family emergency card you can copy and keep in every wallet.

A word on safety. Old ways are not always safe ways. Where tradition and modern science disagree (as with canning times, bleeding powders, or heating a home with fire), this book follows the tested, safer method and tells you why. Watch for the red *Important* notes.

CHAPTERS AT A GLANCE

CHAPTER	WHAT IT COVERS	SKILLS
1. Water	Storing, finding and cleaning the one thing you cannot live without	6
2. Preserving Food	Keeping the harvest the way our great-grandparents did	8
3. The Old-Time Kitchen	Bread, butter and thrift from scratch	7
4. Home Remedies and First Aid	Taking care of your own when help is far away	6
5. The Off-Grid Home	Heat, light, cooling and power when the lines go down	7
6. Garden and Homestead	Growing food on any patch of ground	6
7. Outdoors and Wilderness	Fire, shelter and finding your way	7
8. Safety, Mind and Community	Staying calm, aware and connected	7
9. Fix-It Skills	Mending, sharpening and repairing instead of replacing	6

CHAPTER ONE

1

WATER

Storing, finding and cleaning the one thing you cannot live without







A healthy adult can last weeks without food but only about three days without water. Every plan for independence starts here: how much to store, where to find more, and how to make doubtful water safe to drink. These skills need little money and mostly use things you already own.

IN THIS CHAPTER

01 Storing a Year's Supply of Water
02 Catching Rainwater from Your Roof
03 Making Water Safe: Boil, Bleach, or Sun
04 The Bucket Sand Filter
05 Fresh Water from Seawater
06 Gathering Water from Air and Dew

HOW TO

STORING A YEAR'S SUPPLY OF WATER

SKILL 01 · WATER · 1 WEEKEND · EASY

MATERIALS

- Food-grade containers (HDPE #2 plastic, marked for water)



- Unscented household bleach (6 to 8.25% sodium hypochlorite)



- A funnel or drinking-water hose, tape and marker



- Wooden pallet or boards



Tap water stored in clean, food-grade containers keeps almost indefinitely if it is treated and kept dark. The usual target is one gallon per person per day, with extra for pets, cooking and washing.

1



Work Out Your Total

People x 365 days x 1 gallon. A family of four needs about 1,460 gallons for a full year; start with two weeks (56 gallons) and build up.

2



Wash Every Container

Wash each container with dish soap and hot water, then rinse with a solution of 1 teaspoon bleach per quart of water. Let it air dry.

3

Fill From a Clean Tap

Fill from a clean tap. City-treated water needs nothing more. Well water: add 8 drops of 6% bleach (6 drops of 8.25%) per gallon.



4



Cap, Label and Store

Cap tightly, label with the date filled, and store in a cool, dark place away from fuel, paint and chemicals.

5



Stand Them on Wood

Stand containers on wood, not bare concrete, which can transfer heat and odors.

6



Rotate Twice a Year

Rotate plastic jugs every 6 to 12 months. Use the old water for the garden and refill.

NOTES

A full 55-gallon drum weighs over 450 pounds. Fill it where it will stay.

IMPORTANT

Never reuse milk jugs. The plastic breaks down and milk residue can't be cleaned out.

HOW TO

CATCHING RAINWATER FROM YOUR ROOF

SKILL 02 · WATER · HALF A DAY · EASY

One inch of rain on a 1,000 square-foot roof yields roughly 600 gallons. A simple barrel under a downspout turns that runoff into free water for gardens, flushing, washing and, once purified, drinking.

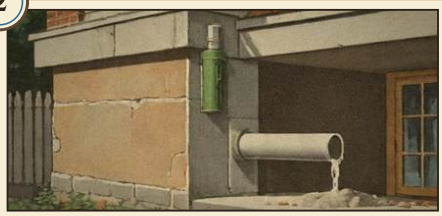
1



Check the Local Rules

Check your local rules first; a few places limit rainwater collection.

2



Stack a Level Stand

Level a spot under a downspout and stack blocks so the barrel sits at least 16 inches high. Height gives you water pressure.

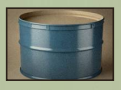
3



Drill the Spigot Hole

Drill a hole near the bottom for the spigot and seal it with washers on both sides.

MATERIALS


A food-grade 55-gallon drum with a lid


Spigot kit with rubber washers


Downspout diverter or flexible elbow


Fine mesh screen (window screen works)


Concrete blocks for a stand


Drill and hole saw

4



Cut and Screen the Inlet

Cut an inlet in the lid and cover it with mesh to keep out leaves and mosquitoes.

5



Add an Overflow Hose

Add an overflow hose near the top that carries extra water away from your foundation.



6 *Connect the Downspout*

Cut the downspout to meet the inlet, or fit a diverter that sends the first dirty flush away.

NOTES

Link two or three barrels with short hoses at the bottom so they fill together.

IMPORTANT

Roof water picks up bird droppings and shingle grit. Filter and disinfect it before drinking.

HOW TO

MAKING WATER SAFE: BOIL, BLEACH, OR SUN

SKILL 03 · WATER · 1 TO 6 HOURS · EASY

MATERIALS

- Pot and heat source, or



- Unscented bleach and a dropper, or



- Clear plastic bottles (PET, 2 liters or smaller)



- Clean cloth for pre-filtering



Clear-looking water can still carry germs. Three methods have been tested for generations and need nothing special.

1



Pre-Filter Through Cloth

Pre-filter: pour cloudy water through a clean cloth or coffee filter, or let it settle and pour off the top.

2



Boil One Full Minute

Boil: bring to a rolling boil for 1 full minute (3 minutes above 6,500 feet). Let cool covered.

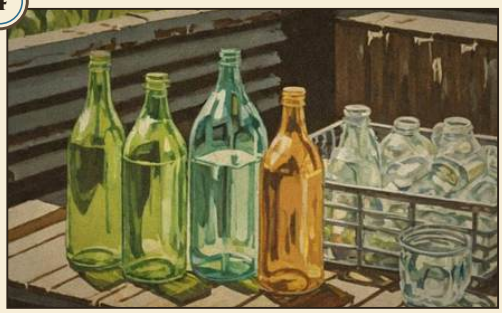
Or Add Bleach by the Drop

Bleach: add 8 drops of 6% bleach (or 6 drops of 8.25%) per gallon. Double it for cloudy water. Stir, wait 30 minutes. It should smell faintly of chlorine; if not, repeat and wait 15 more minutes.

3



4



Or Lay Bottles in the Sun

Sun (Sodis): fill clear bottles with clear water, shake for 20 seconds to add air, and lay them on a dark roof or sheet metal in full sun for 6 hours (2 days if cloudy).

NOTES

To improve the flat taste of boiled water, pour it back and forth between two clean containers a few times.

IMPORTANT

None of these methods removes chemicals, fuel or heavy metals. Avoid water near mines, farms that spray, or roads.

5



Store It Covered

Store treated water in clean, covered containers and drink it within a day or two.

HOW TO

THE BUCKET SAND FILTER

SKILL 04 · WATER · 2 HOURS, THEN ONGOING · MEDIUM

A slow sand filter copies what happens in the ground: water trickles through layers of sand and a living film forms on top that eats germs. It's the same idea used in early city water works.

1



Drill the Buckets

Drill a spigot near the bottom of the lower bucket. Drill many small holes in the bottom of the upper bucket.

2



Rinse Until Clear

Rinse all gravel and sand until the rinse water runs nearly clear.

3



Layer Gravel, Then Sand

Layer the upper bucket from the bottom: 2 inches coarse gravel, 2 inches pea gravel, then 12 to 16 inches of fine sand.

MATERIALS


Two 5-gallon food-grade buckets


Washed fine sand (play sand), about 3 gallons


Pea gravel and coarse gravel, 1 gallon each


Spigot kit


Plate or small dish (to break the fall of the water)


Drill

4



Set a Plate on the Sand

Set a plate on top of the sand so poured water doesn't dig holes.

5



Keep Water Standing on Top

Pour water in and keep about 2 inches standing above the sand at all times. The bio-layer takes 2 to 3 weeks to form.



6 ### *Then Boil or Bleach It*

Run filtered water through boiling or bleach before drinking. The filter makes it clear, the disinfecting makes it safe.

NOTES

When flow slows to a drip, scrape off the top half-inch of sand, rinse it, and put it back.

HOW TO

FRESH WATER FROM SEAWATER

SKILL 05 · WATER · SEVERAL HOURS · MEDIUM

Drinking seawater makes you thirstier and can kill you. But boil it, catch the steam, and the salt stays behind. This works for brackish or polluted water too.

MATERIALS

- Large pot with a lid



- Heat source and fuel



- A clean cup or bowl that fits inside the pot



- Ice or a cold cloth (optional)



1



Pour in a Little Seawater

Pour a few inches of seawater into the pot.

2



Set a Bowl in the Middle

Set the clean bowl in the middle, floating or raised on a stone, so seawater can't slosh in.

3

Lid Upside Down

Put the lid on upside down, so its handle points down over the bowl.



4



Gentle Boil, Steam Drips

Bring to a gentle boil. Steam rises, hits the lid, runs to the low point and drips into the bowl.

5



Ice or a Cold Cloth on Top

Put ice or a wet cold cloth on top of the lid to speed condensing.

6



Taste It: Not Salty

Expect about one cup of fresh water per hour on a camp stove. Taste-test: it should not be salty.

NOTES

Solar version: dig a pit, set a cup in the center, cover with clear plastic weighted with a pebble over the cup. Slow but needs no fuel.

HOW TO

GATHERING WATER FROM AIR AND DEW

SKILL 06 · WATER · OVERNIGHT · EASY

On a still, clear night, grass and metal cool below the air temperature and dew forms. With a cloth and some patience you can collect drinkable water even where there's no stream.

1



Walk the Wet Grass at Dawn

Before sunrise, tie cloths around your ankles and walk through tall wet grass, then wring them into a container. Repeat.

2



Wipe Dew Off Metal and Glass

Wipe dew off car windows, metal roofs or tarps with a clean cloth and wring it out.

3



Bag a Leafy Branch

Tree method: on a sunny day, tie a clear plastic bag over a leafy, non-poisonous branch. Place a pebble in the low corner. Plants sweat water into the bag.

MATERIALS



Absorbent cloths or a cotton t-shirt



A clean container



Plastic sheet or tarp (optional)



Plastic bags (for the tree method)

4



Collect at Day's End

Collect the bag at the end of the day, untie carefully, and pour out the water.

NOTES

Avoid grass that has been sprayed or is near busy roads.



5 ### *Boil It Before Drinking*

Disinfect dew or tree water by boiling or bleach before drinking.

CHAPTER TWO

2

PRESERVING FOOD

Keeping the harvest the way our great-grandparents did







Before refrigerators, every household knew how to keep food through winter. Salt, smoke, fat, dryness, cold and fermentation were the tools. Learn them and you can buy in bulk, waste less, and eat well when the stores are empty.

IN THIS CHAPTER

07 Pemmican: The Original Survival Food
08 Keeping Eggs a Year Without a Fridge
09 Canning Meat in a Jar
10 Rendering Lard and Tallow
11 The Backyard Smokehouse
12 Sun and Air Drying Fruits and Vegetables
13 Sauerkraut and Crock Fermenting
14 The Clay Pot Refrigerator

HOW TO

PEMMICAN: THE ORIGINAL SURVIVAL FOOD

SKILL 07 · PRESERVING FOOD · 2 DAYS · MEDIUM

Plains peoples of North America made pemmican from dried meat and rendered fat. Properly made, it keeps for years without refrigeration and carries more energy per pound than nearly any other food.

MATERIALS

- 2 lb very lean beef, venison or bison



- About 1 lb rendered beef tallow (see Rendering Lard and Tallow)



- Dried berries (optional, unsweetened)



- Oven or dehydrator



- Blender or mortar and pestle



- Baking pan and parchment



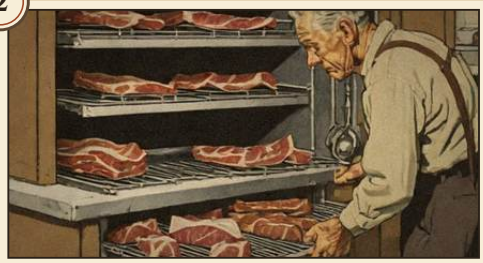
1



Trim and Slice Thin

Trim every bit of fat from the meat and slice it as thin as possible.

2



Dry Until It Snaps

Dry at 120 to 170 F until it snaps like a cracker, not bends. This can take 8 to 16 hours.

3

Grind to a Powder

Grind the dried meat to a fine powder. Grind berries separately and mix in.



4



Melt the Tallow Gently

Gently melt the tallow until liquid but not smoking.

5



Mix Equal Weights

Mix equal weights of meat powder and melted tallow. It should look like moist sand that just holds together.

6



Press, Cool and Cut

Press into a pan, cool until firm, and cut into bars. Wrap in wax paper and store cool and dark.

NOTES

Eat it plain, fry it, or crumble it into boiling water with a potato for a hearty stew.

IMPORTANT

Any moisture or lean fat left in will spoil it. Dry the meat completely.

HOW TO

KEEPING EGGS A YEAR WITHOUT A FRIDGE

SKILL 08 · PRESERVING FOOD · 30 MINUTES · EASY

'Water glassing' was common on farms a century ago. Fresh, unwashed eggs sealed in a lime solution stay usable for 8 to 12 months at cool room temperature.

MATERIALS


Fresh, clean, unwashed eggs with no cracks


Pickling lime (calcium hydroxide)


Filtered water


Glass jar or food-grade bucket with lid


Kitchen scale

1



Unwashed Eggs Only

Use only eggs that have never been washed; washing removes the natural 'bloom' coating that seals the shell.

2



Mix Lime Into Water

Mix 1 ounce (by weight) of pickling lime into each quart of water. Stir until mostly dissolved.

3



Point End Down

Gently place eggs in the jar, pointy end down. Do not drop them.

4



Cover With Lime Water

Pour lime water over them until covered by at least an inch.

5



Seal and Store Cool

Seal and store between 40 and 70 F in a dark spot.



6 *Rinse and Sniff Test*

To use, rinse each egg well. Crack into a cup first and sniff to check it's good.

NOTES

Shells may be weaker after storage. Water-glassed eggs are best for scrambling and baking; they may not whip well.

IMPORTANT

Pickling lime is caustic. Wear gloves and keep it away from eyes.

HOW TO

CANNING MEAT IN A JAR

SKILL 09 · PRESERVING FOOD · 3 HOURS · MEDIUM

Home-canned ground meat, stew beef or chicken is ready to eat and keeps on a shelf for years. Meat is a low-acid food, so it MUST be pressure canned; a water bath is not enough.

1



Brown the Meat Lightly

Brown ground meat lightly and drain the fat. Raw chunks can go in raw.

2



Pack Hot Jars

Pack into hot jars leaving 1 inch of headspace. Add 1/2 teaspoon salt per pint if you like.

3

Add Broth, Free the Bubbles

Add boiling water or broth to keep 1 inch of headspace. Remove bubbles with a plastic knife.



4



Wipe Rims, Lids Fingertip Tight

Wipe rims with a clean cloth dipped in vinegar. Fat on the rim stops a seal. Add lids and rings fingertip tight.

5



Process at Pressure

Process at the pressure for your altitude. For ground meat at sea level (dial gauge): 11 pounds, 75 minutes for pints, 90 for quarts. Check your guide.

6



Cool and Check the Seal

Let pressure drop by itself. Remove jars, cool 12 to 24 hours, and check that lids don't flex.

NOTES

Store without the rings so a failed seal is easy to spot.

IMPORTANT

Botulism can't be seen, smelled or tasted. Follow tested times exactly. Never can meat in an oven or open kettle. If a lid bulges or spurts, throw it out.

MATERIALS

- A pressure canner with a working gauge



- Pint or quart canning jars, new lids and rings



- Ground beef, chunks of meat, or chicken



- Salt (optional)



- Jar lifter, funnel, clean towels



- A current tested guide (USDA or university extension)



HOW TO

RENDERING LARD AND TALLOW

SKILL 10 · PRESERVING FOOD · 4 TO 6 HOURS · EASY

Pork fat becomes lard; beef or mutton fat becomes tallow. Both are prized for frying, baking, soap, candles and preserving. Many butchers give fat away for little or nothing.

MATERIALS


Pork back fat or leaf fat, or beef suet, 3 to 5 lb


Heavy pot or slow cooker


Cheesecloth or a fine strainer


Clean glass jars


A little water

1



Cube the Chilled Fat

Chill the fat so it's firm, then cut into small cubes or run through a grinder.

2



A Splash of Water First

Put it in the pot with 1/4 cup water to stop scorching at the start.

3



Heat Low and Stir

Heat on low. Stir every 20 minutes. Fat will melt and solid bits ('cracklings') will brown and float.

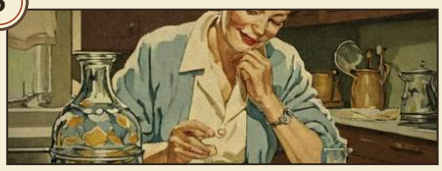
4



Cracklings Sink: Done

When cracklings sink and bubbling slows, the fat is done. Do not let it smoke.

5



Strain Into Warm Jars

Strain through cheesecloth into warm, clean jars.



6 *Cool and Cap*

Cool and cap. Lard keeps months in a cool pantry, a year in the fridge, longer frozen.

NOTES

Salt the cracklings and eat them as a snack. Leaf lard (around the kidneys) makes the whitest, mildest lard for pie crust.

HOW TO

THE BACKYARD SMOKEHOUSE

SKILL II · PRESERVING FOOD · 1 TO 2 WEEKENDS · HARD

Cold smoke dries and flavors meat and fish; hot smoke cooks it. A small smokehouse with the fire set apart lets you do both and was found on most farms before 1940.

1



Pick a Sloping Spot

Pick a spot on a slight slope. The firebox goes at the bottom, the smokehouse uphill.

2



Build a Tight Box

Build the box with a tight door, a few small vents near the top, and hooks or racks inside.

3

Lay the Pipe in a Trench

Lay the pipe in a shallow trench from firebox to the smokehouse floor. The distance cools the smoke for cold smoking.



4



Cure the Meat First

Cure meat first with salt (and sugar or spices) in the fridge for the time your recipe gives.

5



Cold Smoke or Hot Smoke

Cold smoke below 90 F for flavor and drying (bacon, ham, cheese, fish). Hot smoke at 160 to 225 F to cook.

6



Thin Blue Smoke, Not White

Keep a thin blue smoke. Thick white smoke makes food bitter.

NOTES

Cold-smoked meat is NOT cooked. Cook it before eating unless it was properly cured and dried by a tested method.

MATERIALS

- Lumber or plywood for a box about 3 x 3 x 6 feet



- Metal roof sheet



- Stovepipe or clay pipe, 10 feet



- Firebox (old wood stove, steel drum or stone pit)



- Hooks, racks, a thermometer



- Hardwood: hickory, apple, oak, maple



HOW TO

SUN AND AIR DRYING FRUITS AND VEGETABLES

SKILL 12 · PRESERVING FOOD · 2 TO 4 DAYS · EASY

Drying removes the water that germs need. In hot, dry weather the sun does all the work; elsewhere a low oven or screen rack near a wood stove does the job.

MATERIALS



Ripe produce (apples, tomatoes, peppers, beans, herbs)



Window screens or a wooden frame with food-safe mesh



Cheesecloth to cover



Lemon juice



Glass jars

1



Slice Thin and Even

Slice evenly, about 1/4 inch thick. Dip apples and pears in lemon water to slow browning.

2



One Layer, Not Touching

Lay in a single layer on the screen, not touching.

3



Full Sun, Off the Ground

Set the rack in full sun, raised off the ground for airflow. Cover with cheesecloth against insects.

4



Bring It In at Night

Bring trays in at night so dew doesn't wet them.

5



Leathery, No Moist Spots

Fruit is done when leathery with no moist spots; vegetables when brittle.



6 *Condition in a Jar*

Condition: pack loosely in a jar for a week, shaking daily. If moisture beads appear, dry longer.

NOTES

String green beans on thread and hang them in a warm attic. Mountain families called them 'leather britches'.

HOW TO

SAUERKRAUT AND CROCK FERMENTING

SKILL 13 · PRESERVING FOOD · 20 MINUTES, THEN 3 WEEKS · EASY

Lacto-fermentation needs only vegetables and salt. Good bacteria turn the sugars into acid, which preserves the food and adds a sharp, rich flavor. One head of cabbage makes about a quart.

1



Core and Shred Finely

Remove outer leaves. Core and shred the cabbage finely.

2



Salt It and Squeeze

Toss with salt in a big bowl and squeeze hard with your hands for 5 to 10 minutes until it's limp and swimming in its own brine.

3

Pack Until Brine Rises

Pack tightly into the crock, pressing down until brine rises over the top.



4



Leaf and Weight on Top

Cover with a whole leaf, then the weight. Everything must stay under the brine.

5



Cover and Wait

Cover with a cloth and leave at 60 to 70 F. Taste after a week; most like it at 3 to 4 weeks.

6



Move It Somewhere Cool

Move to a cool cellar or fridge when it tastes right.

NOTES

A white film on top (kahm yeast) can be skimmed off. Fuzzy colored mold means throw it out.

MATERIALS

- 2 heads green cabbage (about 5 lb)



- 3 tablespoons non-iodized salt



- A crock or wide-mouth jars



- A weight (small jar of water or clean stone)



- Cloth cover



HOW TO

THE CLAY POT REFRIGERATOR

SKILL 14 · PRESERVING FOOD · 1 HOUR · EASY

The 'zeer' pot uses evaporation to keep vegetables 15 to 20 degrees cooler than the air. It's widely used in hot, dry places and needs no power at all.

MATERIALS



Two unglazed clay pots, one about 2 inches smaller all round



Sand to fill the gap



Water



A wet cloth or lid



Cork or clay to plug drain holes

1



Plug the Drain Holes

Plug the drain holes of both pots.

2



Pot Inside Pot

Put an inch of sand in the large pot and set the smaller pot inside it.

3



Fill the Gap With Sand

Fill the gap between them with sand.

4



Soak the Sand

Soak the sand with water until it's damp all the way through.

5



Load and Cover With a Wet Cloth

Put vegetables in the inner pot and cover with a wet cloth.



6 ### *Shade and a Breeze*

Keep it in a shady, breezy spot. Re-wet the sand once or twice a day.

NOTES

It works best when the air is dry. In humid weather, a root cellar or spring house works better.

CHAPTER THREE

3

THE OLD-TIME KITCHEN

Bread, butter and thrift from scratch







Our grandparents cooked from basic staples: flour, fat, salt, beans and whatever the garden gave. These recipes and habits stretch a pantry a long way and taste better than anything from a box.

IN THIS CHAPTER

15 Yeast-Free Soda Bread
16 Starting and Keeping Sourdough
17 Hardtack: Bread That Lasts Decades
18 Depression-Era Meals That Stretch a Dollar
19 Cast-Iron Care (and the Mistake Everyone Makes)
20 Churning Butter in a Jar
21 Twelve Uses for Eggshells

HOW TO

YEAST-FREE SODA BREAD

SKILL 15 · THE OLD-TIME KITCHEN · 1 HOUR · EASY

MATERIALS

- 4 cups flour



- 1 teaspoon baking soda



- 1 teaspoon salt



- 1 3/4 cups buttermilk (or milk with 1 tablespoon vinegar)



- Mixing bowl, baking sheet or cast-iron pan



When yeast runs out, baking soda and sour milk make a fine loaf. Irish farm wives baked this daily in iron pots over open fires.

1



Heat the Oven to 425 F

Heat the oven to 425 F.

2



Whisk the Dry Things

Whisk flour, soda and salt in a bowl.

3

Stir in the Buttermilk

Pour in most of the buttermilk and stir with a fork until a shaggy dough forms. Add more milk if dry.



4



Shape a Round, Barely Knead

Turn out and shape into a round with floured hands. Do not knead much; it gets tough.

5



Cut a Deep Cross

Cut a deep cross in the top 'to let the fairies out' (and the heat in).

6



Bake Until It Sounds Hollow

Bake 35 to 40 minutes until it sounds hollow when tapped underneath.

NOTES

In a Dutch oven over coals: preheat the pot, add the loaf, put the lid on and pile coals on top. Check at 30 minutes.

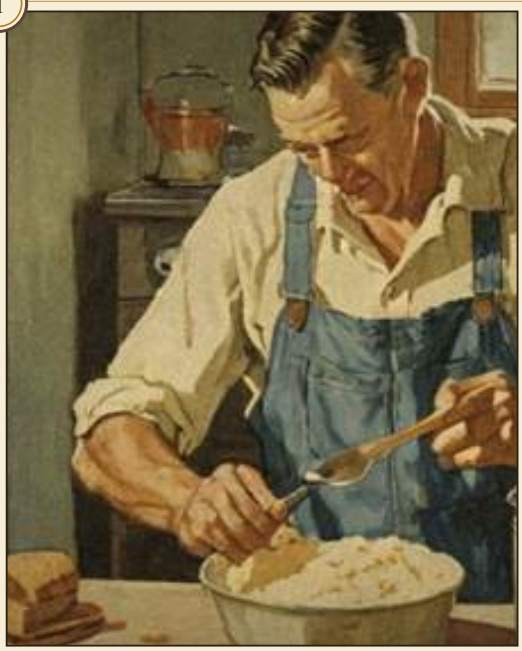
HOW TO

STARTING AND KEEPING SOURDOUGH

SKILL 16 · THE OLD-TIME KITCHEN · 7 DAYS TO START · MEDIUM

A sourdough starter is wild yeast you catch from the air and flour. Kept fed, it will raise bread for a lifetime; some families pass theirs down for a century.

1



Day 1: Flour and Water

Day 1: Mix 50 g flour and 50 g water in the jar. Cover with cloth. Leave at room temperature.

2



Feed It Every Day

Days 2 to 7: Each day, throw out half and add 50 g flour and 50 g water. Stir well.

3



By Day 7, It Doubles

By day 5 to 7 it should double within 6 to 8 hours of feeding and smell pleasantly sour.

MATERIALS



Whole wheat or rye flour, then any flour



Unchlorinated water (let tap water sit out overnight)



A quart jar and a cloth cover



Kitchen scale (helpful)

4



Use It at the Peak

To bake, use it at its peak, when bubbly and domed.

5



Rest in the Fridge

Keep it in the fridge between bakes and feed once a week.

NOTES

A layer of gray liquid ('hooch') means it's hungry. Pour it off and feed.



6 ### *Dry Some for Backup*

For backup, smear some thin on parchment, dry it, and crumble into a jar. It revives with flour and water.

HOW TO

HARDTACK: BREAD THAT LASTS DECADES

SKILL 17 · THE OLD-TIME KITCHEN · 1 HOUR · EASY

Sailors and soldiers lived on these rock-hard crackers. Kept dry, hardtack lasts years. It's not exciting, but it is filling and nearly indestructible.

MATERIALS

- 4 cups flour



- 1 1/2 to 2 cups water



- 2 teaspoons salt



- Rolling pin, knife, a nail or skewer



- Baking sheet



1



Heat the Oven to 375 F

Heat oven to 375 F.

2



A Stiff Dough

Mix flour and salt, then add water a little at a time until it forms a stiff dough that doesn't stick.

3

Roll Out and Cut Squares

Roll out to 1/2 inch thick. Cut into 3-inch squares.



4



Poke 16 Holes

Poke 16 holes through each square with a skewer.

5



Bake, Flip, Bake Again

Bake 30 minutes, flip, and bake 30 more. Then leave in the turned-off oven to dry completely.

6



Store in an Airtight Tin

Store in an airtight tin.

NOTES

Soak in coffee, soup or water before eating, or crumble and fry in bacon fat with onions, a Civil War dish called 'skillygalee'.

HOW TO

DEPRESSION-ERA MEALS THAT STRETCH A DOLLAR

SKILL 18 · THE OLD-TIME KITCHEN · VARIES · EASY

In the 1930s, families fed six on what one person eats today. These meals use cheap staples and waste nothing.

1



Potato Soup

Fry an onion in grease, add diced potatoes and water to cover, simmer soft, mash half, and thin with milk. Salt and pepper.

2



Beans and Cornbread

Soak dry beans overnight, simmer with a ham bone or bacon end 2 to 3 hours. Serve with skillet cornbread.

3



Milk Toast

Stale bread toasted, buttered and covered with hot sweetened milk.

MATERIALS



Potatoes, onions, beans, flour, oats, cornmeal



Bacon grease or lard



Stale bread



Leftover scraps and bones

4



Fried Mush

Cook cornmeal porridge, chill it in a loaf pan overnight, slice and fry crisp.

5



Stone Soup Broth

Stone Soup: save vegetable peels and bones in a jar in the freezer; when full, simmer 3 hours for broth.

NOTES

Rice and beans together make a complete protein. A 20-lb bag of each feeds a family for weeks.



6 ### *Bread Pudding*

Bake bread pudding from any stale bread, eggs, milk and a little sugar.

HOW TO

CAST-IRON CARE (AND THE MISTAKE EVERYONE MAKES)

SKILL 19 · THE OLD-TIME KITCHEN · 1 HOUR · EASY

A cast-iron skillet properly cared for will outlive you. The common mistake is letting it soak or drip dry, which brings on rust. The fix is quick heat and a thin coat of oil.

MATERIALS

- Cast-iron pan



- Stiff brush or chain-mail scrubber



- Coarse salt



- A high-smoke-point fat (lard, shortening or vegetable oil)



- Paper towel or rag



1



Clean It While Warm

Clean while still warm: scrape out food, scrub with hot water. A little mild soap is fine now; old lye soaps were the problem.

2



Scrub With Coarse Salt

For stuck bits, scrub with coarse salt and a little oil.

3

Dry It on the Burner

Dry right away with a towel, then set on the burner for a minute or two to drive off the last water.



4



A Few Drops of Fat

Rub in a few drops of fat, then wipe until it looks almost dry. Too much oil gets sticky.

5



Re-Season a Rusty Pan

To re-season a rusty pan: scrub to bare metal, oil thinly all over, bake upside down at 450 F for an hour. Repeat 3 times.

6



Store Dry With Paper Inside

Store in a dry place with a paper towel inside.

NOTES

Avoid long simmering of tomatoes or vinegar in a new pan; acid strips the seasoning.

HOW TO

CHURNING BUTTER IN A JAR

SKILL 20 · THE OLD-TIME KITCHEN · 20 MINUTES · EASY

Fresh butter needs nothing but cream and elbow grease. The buttermilk left over is perfect for soda bread.

1



Half a Jar of Cream

Fill the jar halfway with cream and seal.

2



Shake Hard

Shake hard. In about 5 minutes it turns to whipped cream; keep going.

3



It Sloshes: Butter!

Suddenly, it will slosh. Yellow butter has separated from thin buttermilk.

MATERIALS



1 pint heavy cream (at cool room temperature)



A quart jar with a tight lid, or a mixer



Ice water



A wooden spoon



Salt (optional)

4



Pour Off the Buttermilk

Pour off and save the buttermilk.

5



Rinse With Ice Water

Rinse the butter with ice water, pressing with the spoon, until the water runs clear. This keeps it from going sour.

NOTES

A French butter bell keeps butter soft and fresh on the counter for weeks by sealing it under a little water.



6 *Salt and Pack*

Work in 1/4 teaspoon salt if you like. Pack into a dish.

HOW TO

TWELVE USES FOR EGGSHELLS

SKILL 21 · THE OLD-TIME KITCHEN · MINUTES · EASY

MATERIALS

- Rinsed eggshells



- A baking sheet



- A blender or mortar and pestle



- A jar



Eggshells are nearly pure calcium carbonate. Rather than throw them out, dry them and put them to work.

1



Rinse, Dry and Sterilize

Rinse shells and dry them on a tray; bake 10 minutes at 200 F to sterilize.

2



Crush or Grind

Crush coarse for the garden, or grind to powder for other uses.

3

In the Garden

Garden: scatter around tomatoes and peppers for calcium; use halves as seed-starting pots; ring seedlings to deter slugs.



4



For the Hens

Chickens: feed crushed shells back to laying hens for stronger shells.

5



Around the House

House: use as a gentle scouring powder for pots; drop a few in a coffee pot to cut bitterness; shake with water in a vase to scrub it.

6



Into the Compost

Compost: they add minerals and lighten the pile.

NOTES

Mix shell powder with a little soap for a non-scratch cleanser for stained mugs.

CHAPTER FOUR

4

HOME REMEDIES AND FIRST AID

Taking care of your own when help is far away







When the roads are closed or the doctor is an hour away, the first minutes matter most. This chapter covers basic first aid and the gentle kitchen remedies families have used for colds and scrapes. None of it replaces a doctor; it helps you bridge the gap until you reach one.

IN THIS CHAPTER

22 Stopping Serious Bleeding
23 Building a Home Medical Kit
24 Oral Rehydration Solution
25 Fire Cider and Honey Garlic
26 The Plantain Poultice
27 Elderberry Syrup

STOPPING SERIOUS BLEEDING

SKILL 22 · HOME REMEDIES AND FIRST AID · MINUTES · EASY

Uncontrolled bleeding can kill in less than five minutes. Pressure, packing and, for limbs, a tourniquet are the tools. Take a hands-on 'Stop the Bleed' class; they're often free.

MATERIALS



Gloves if you have them



Clean cloth, gauze or hemostatic gauze



Commercial tourniquet (CAT or SOF-T style)



Phone to call 911

1



Call 911, Protect Yourself

Call 911 or send someone to call. Protect yourself with gloves or a plastic bag.

2



Press Hard on the Wound

Find the source of the bleeding. Press hard on it with cloth and both hands. Don't lift to check.

3



Pack a Deep Wound

For a deep wound, pack it: stuff gauze into the wound tight against the bleeding, then press for at least 3 minutes.

4



Tourniquet Above the Wound

If a limb wound spurts or won't stop, apply a tourniquet 2 to 3 inches above the wound (not on a joint). Tighten until bleeding stops. It will hurt.

5



Note the Time

Note the time it went on and write it on the person's skin or tag.



6 ### *Keep Them Warm and Still*

Keep the person warm and lying down. Don't remove the tourniquet; leave that to medical staff.

NOTES

IMPORTANT

Old-time 'stop bleeding' powders like flour, cobwebs or ground pepper can cause infection. Use clean pressure. Modern clotting gauze is the safe version of that old idea.

HOW TO

BUILDING A HOME MEDICAL KIT

SKILL 23 · HOME REMEDIES AND FIRST AID · 1 AFTERNOON · EASY

A well-stocked kit in a known spot saves time when it counts. Build it in a sturdy tackle box or tool bag and check it twice a year.

1



Wound Care

Wounds: assorted bandages, gauze pads and rolls, medical tape, elastic wrap, butterfly closures, antibiotic ointment, saline.

2



Trauma

Tourniquet, pressure bandage, hemostatic gauze, trauma shears, gloves.

Medicines

Pain/fever reducer, antihistamine, anti-diarrhea tablets, oral rehydration salts, hydrocortisone cream, and a 2-week supply of everyone's prescriptions if you can get it.

3



4



Tools

Tweezers, thermometer, splint, flashlight, safety pins, instant cold packs, emergency blanket.

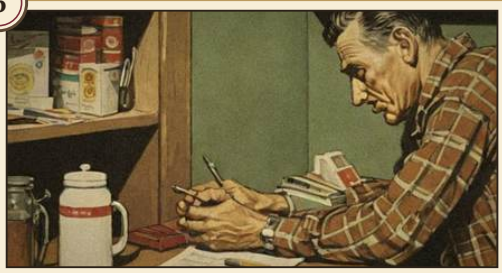
5



Paperwork

A card for each person with allergies, medicines, blood type and doctor's number.

6



Check Dates Twice a Year

Mark a calendar to check expiration dates every spring and fall.

NOTES

Keep a smaller copy in each car and in your go-bag.

MATERIALS

- A clearly labeled waterproof box



- The list below



- A first-aid manual



HOW TO

ORAL REHYDRATION SOLUTION

SKILL 24 · HOME REMEDIES AND FIRST AID · 5 MINUTES · EASY

Diarrhea and vomiting kill through dehydration. A simple sugar-salt drink, promoted worldwide by health agencies, has saved millions of lives.

MATERIALS


1 liter (about 1 quart) clean, safe water


6 level teaspoons sugar


1/2 level teaspoon salt


A clean jug and spoon

1



Measure Carefully

Measure carefully. Too much salt is harmful, especially for children.

2



Dissolve Completely

Dissolve sugar and salt completely in the water.

3



Small Sips, Often

Give small sips often: a spoonful every minute or two for a child, more for adults.

4



Keep It Up

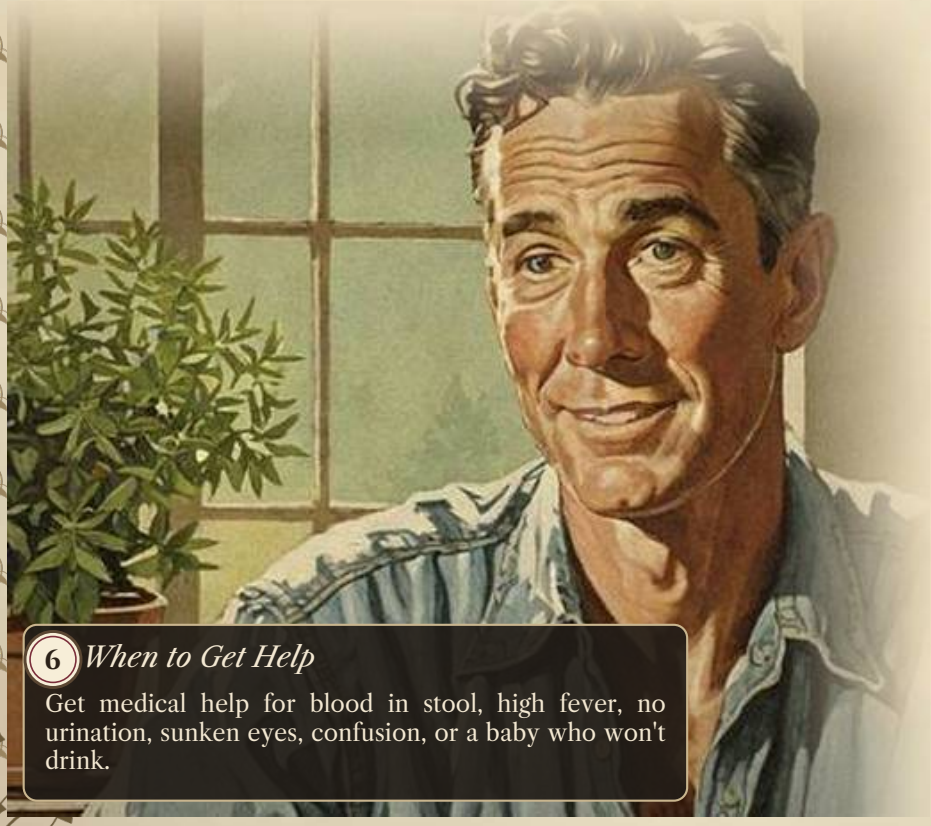
Keep giving it as long as diarrhea lasts. Keep breastfeeding babies.

5



Fresh Batch Every Day

Make a fresh batch each day.



6 *When to Get Help*

Get medical help for blood in stool, high fever, no urination, sunken eyes, confusion, or a baby who won't drink.

NOTES

A squeeze of orange or a mashed banana adds potassium and helps flavor.

HOW TO

FIRE CIDER AND HONEY GARLIC

SKILL 25 · HOME REMEDIES AND FIRST AID · 20 MINUTES, THEN WEEKS · EASY

Grandmothers swore by these pantry tonics in cold season. They taste bold, they're comforting for a sore throat, and they make a fine salad dressing too. They're a food, not a cure.

MATERIALS

- Raw honey



- Fresh garlic cloves



- Apple cider vinegar



- Onion, horseradish, fresh ginger, hot pepper, lemon



- Clean glass jars with plastic lids



1



Honey Garlic: Crush and Cover

Honey Garlic: peel and lightly crush cloves, fill a jar halfway, cover with honey. Leave the lid loose; 'burp' and flip it daily for the first weeks as it bubbles.

2



Ready in a Month

It's ready in about a month. Take a spoonful in tea or eat a clove.

3

Fire Cider: Chop It All

Fire Cider: fill a jar with chopped onion, garlic, horseradish, ginger, a hot pepper and lemon slices.



4



Cover With Vinegar

Cover with apple cider vinegar. Use a plastic lid or parchment under metal, since vinegar corrodes it.

5



Steep, Shake, Strain

Steep 3 to 4 weeks in a dark cupboard, shaking daily. Strain and sweeten with honey to taste.

6



A Spoonful in Warm Water

Take a tablespoon in warm water, or use as a dressing.

NOTES

IMPORTANT

Never give honey to babies under 1 year. Garlic in oil is a botulism risk; honey garlic should stay acidic and fermenting. Throw it out if it smells off or grows mold.

HOW TO

THE PLANTAIN POULTICE

SKILL 26 · HOME REMEDIES AND FIRST AID · 5 MINUTES · EASY

Broadleaf plantain (not the banana) grows in nearly every lawn. Herbalists have long used it on bee stings, bug bites and minor scrapes.

MATERIALS


A few fresh plantain leaves from an unsprayed yard


Clean water


A bandage or cloth strip

1



Learn the Plant

Low rosette of oval leaves with 5 to 7 strong parallel veins running stem to tip; skinny seed spikes in summer.

2



Pick Clean Leaves

Pick a few clean leaves away from roads and sprayed areas.

3



Crush Until Juicy

Rinse, then chew or crush the leaf until juicy (the 'spit poultice').

4



Press Onto the Sting

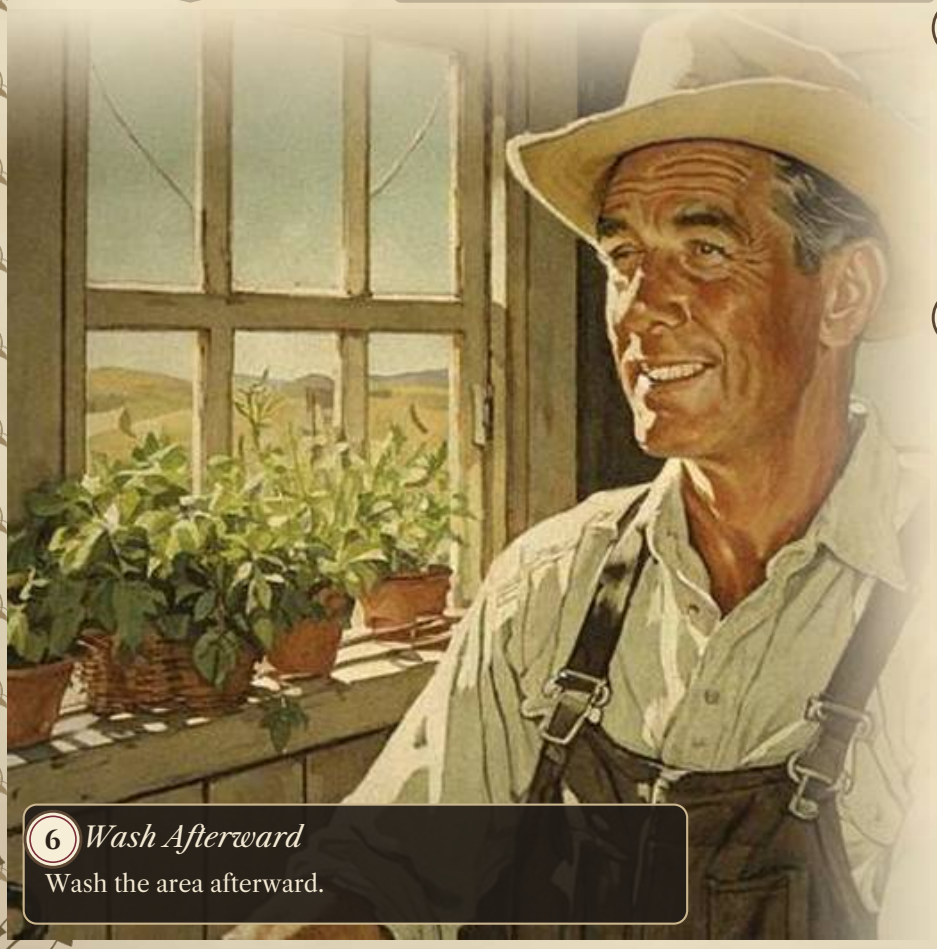
Press the mash onto the sting or bite.

5



Bandage for 20 to 30 Minutes

Hold it in place with a bandage for 20 to 30 minutes. Replace with fresh leaves if needed.



6 *Wash Afterward*

Wash the area afterward.

NOTES

Remove a bee stinger first by scraping sideways with a card, not squeezing.

IMPORTANT

Get medical help at once for any swelling of face or throat or trouble breathing after a sting.

HOW TO

ELDERBERRY SYRUP

SKILL 27 · HOME REMEDIES AND FIRST AID · 1 HOUR · EASY

Black elderberry has been a folk staple in Europe and America for centuries. The syrup is a tasty winter tradition.

MATERIALS

- 2/3 cup dried black elderberries (Sambucus nigra) or 1 1/3 cups fresh, ripe



- 3 1/2 cups water



- 1 cinnamon stick, a few cloves, sliced ginger



- 1 cup raw honey



- Strainer and a glass bottle



1



Berries, Water and Spices

Combine berries, water and spices in a pot.

2



Simmer Until Reduced by Half

Bring to a boil, then simmer uncovered about 45 minutes until reduced by half.

3

Mash and Strain

Mash the berries, then strain through a fine strainer, pressing hard.



4



Stir in the Honey, Warm Not Hot

Cool until warm, not hot, and stir in the honey.

5



Bottle and Refrigerate

Bottle and keep in the fridge up to 2 months.

6



A Spoonful a Day

Usual amount: 1 tablespoon a day for adults; 1 teaspoon for children over 1.

NOTES

IMPORTANT

Raw elderberries, stems and leaves contain compounds that cause nausea. Always cook berries, and only use ripe, dark fruit. No honey for babies.

CHAPTER FIVE

5

THE OFF-GRID HOME

Heat, light, cooling and power when the lines go down







Ice storms, heat waves and hurricanes can knock out power for days or weeks. A few low-tech skills keep a home livable, and they cost far less than a whole-house generator.

IN THIS CHAPTER

28 Cooling a House Without Electricity
29 Staying Warm in a Winter Blackout
30 The Rocket Stove
31 Light Without Power
32 Ten Uses for Wood Ash
33 A Trash-Can Faraday Cage
34 Simple Solar Backup Power

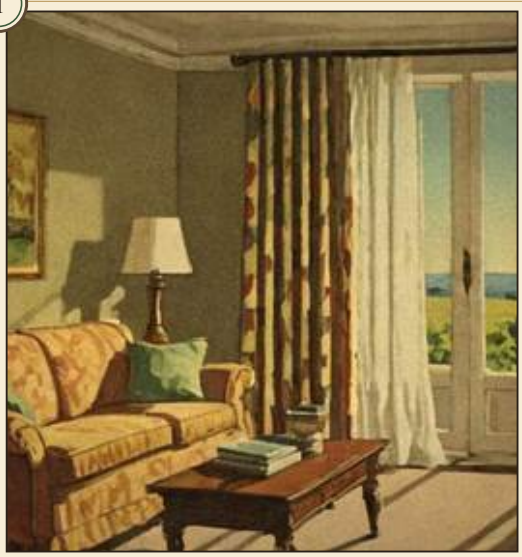
HOW TO

COOLING A HOUSE WITHOUT ELECTRICITY

SKILL 28 · THE OFF-GRID HOME · ONGOING · EASY

Before air conditioning, houses stayed bearable through smart habits: shade, cross-breezes and night cooling.

1



Shut Out the Morning Sun

Close windows and curtains on the sunny side by 9 a.m. Heat that never enters doesn't need removing.

2



Open Low and High at Night

At night, open windows low on the cool side and high on the warm side. Hot air rises out and cool air is pulled in.

3



Hang a Damp Sheet

Hang a damp sheet in an open window. The breeze through it cools by evaporation.

MATERIALS



Window coverings (blankets, foil-backed shades)



Spray bottle and damp cloths



Battery or solar fans if available

4



Cook Outdoors

Cook outdoors or eat cold meals so the stove doesn't heat the house.

5



Go Low

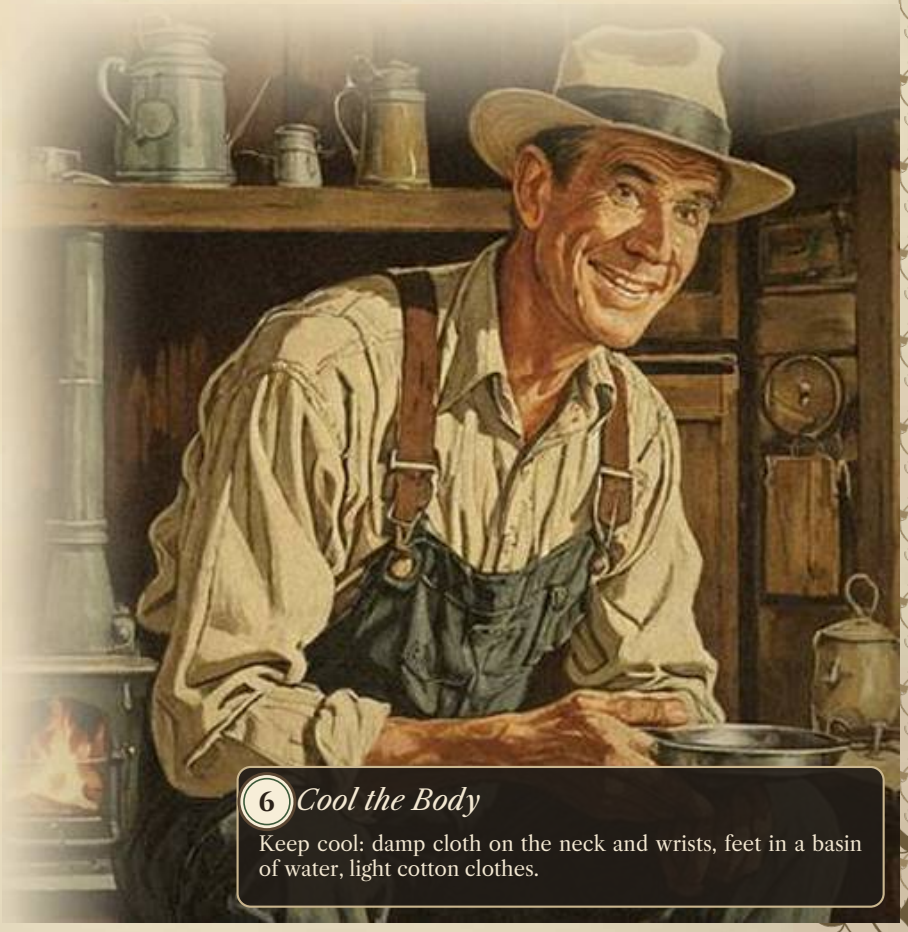
Spend the hottest hours in the lowest room; basements stay coolest.

NOTES

Plant deciduous trees on the south and west sides. They shade in summer and let sun through in winter.

IMPORTANT

Learn heatstroke signs: hot dry skin, confusion, fast pulse. It's an emergency. Cool the person and call for help.



6 *Cool the Body*

Keep cool: damp cloth on the neck and wrists, feet in a basin of water, light cotton clothes.

HOW TO

STAYING WARM IN A WINTER BLACKOUT

SKILL 29 · THE OFF-GRID HOME · ONGOING · EASY

The trick is to heat people, not the whole house. Pick one small room and make it warm.

MATERIALS

- Blankets, sleeping bags, wool layers



- Battery carbon-monoxide alarm



- Indoor-safe heater (properly vented wood stove, or one rated for indoor use)



- Tent (optional)



1



Pick One Small Room

Choose a small interior room with few windows. Close off the rest of the house.

2



Seal Windows and Doors

Cover windows with blankets or bubble wrap, and stuff towels under doors.

3

Tent Up Indoors

Set up a tent indoors on a rug and sleep in it together. Body heat in a small space adds up quickly.



4



Dress in Layers

Wool or synthetic next to skin, then fleece, then a windproof layer. Wear a hat to bed.

5



Warm Food, Hot Bottles

Eat and drink warm things. Fill hot water bottles or bottles of hot water for sleeping bags.

6



Keep the Pipes From Freezing

Drip faucets and open cabinet doors under sinks to keep pipes from freezing.

NOTES

IMPORTANT

Never use a grill, camp stove, gas oven or generator indoors or in a garage. Carbon monoxide kills silently. Keep generators 20 feet from the house, and run a CO alarm.

HOW TO

THE ROCKET STOVE

SKILL 30 · THE OFF-GRID HOME · 1 HOUR · MEDIUM

A rocket stove burns small sticks so hot and clean that it can boil water with a handful of twigs. You can build one from cans or bricks.

1



Four Bricks, One Gap

Brick Version: lay 4 bricks in a square leaving a gap at the front for feeding wood. This is level one.

2



Stack Five Levels

Stack 4 more levels, staggering joints, leaving the feed hole open only at the bottom.

3



A Rod for the Sticks

Place a metal rod across the feed hole so sticks rest on it and air flows under them.

MATERIALS



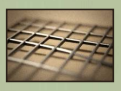
16 bricks, or a large #10 can and 4 soup cans



Tin snips (for the can version)



Sand, wood ash or perlite for insulation



A grate or a few metal rods

4



Pot on Three Stones

Set the pot on three small stones or bricks on top so flames and smoke can escape.

5



Light It, Feed It Sideways

Light paper and twigs at the opening, then feed thumb-thick sticks in from the side.

NOTES

The 'L' shape is the secret: the vertical chimney draws air through the fire like a rocket nozzle.



6 ### *The Tin-Can Version*

Can Version: make an L-shaped chimney from soup cans inside the big can and pack ash around it for insulation.

HOW TO

LIGHT WITHOUT POWER

SKILL 31 · THE OFF-GRID HOME · MINUTES · EASY

MATERIALS

- Headlamps and lanterns with spare batteries



- Kerosene or oil lamp with wicks and fuel



- Long-burning candles and holders



- Solar garden lights



Flashlights fail; batteries run out. Old-fashioned light sources keep working, and a few are very cheap to stock.

1



A Headlamp for Everyone

Give everyone a headlamp; it frees both hands.

2



Solar Lights Come Inside

Bring solar garden lights inside at night and put them back in the sun by day.

3

Oil Lamp: Trim, Fill, Light

Oil lamp: trim the wick straight, fill the font no more than 3/4, light with the chimney off, then replace it and turn the wick down until there's no smoke.



4



Emergency Jar Lamp

Emergency lamp: a jar of vegetable oil with a cotton wick held by a wire clip burns for hours.

5



Candles Safely

Place candles in a deep jar or on a plate, away from curtains, and never leave them burning when you sleep.

6



A Mirror Doubles the Light

Put a mirror behind a lamp to double the light in the room.

NOTES

A headlamp strapped to a gallon jug of water, facing in, turns it into a glowing lantern that lights a whole room.

HOW TO

TEN USES FOR WOOD ASH

SKILL 32 · THE OFF-GRID HOME · MINUTES · EASY

If you burn wood, you make ash. Hardwood ash is alkaline and rich in potassium, and pioneers found a dozen uses for it.

1



Store Ash in Metal

Safety: store ash in a metal can with a lid, away from the house. Coals can hide in ash for days.

2



Sweeten the Garden Soil

Garden: sprinkle lightly on acid soil to raise pH and add potassium. Skip it near blueberries and potatoes.

3



A Ring Against Slugs

Slugs: ring tender plants with a band of dry ash.

MATERIALS



Cold hardwood ash, sifted



Metal bucket with a lid



A dry place to store it

4



Grip on Icy Steps

ICE: scatter on icy steps for grip.

5



Scour Glass and Tarnish

Cleaning: a damp paste of ash scours glass stove doors and tarnish. Wear gloves.

NOTES

IMPORTANT

Lye from ash is caustic and can blind. Only make soap with gloves, goggles and a tested recipe.



6 ### *Lye for Soap: Full Protection*

Soap: leach water through ash for lye, the old base of homemade soap (handle with full eye and skin protection).

HOW TO

A TRASH-CAN FARADAY CAGE

SKILL 33 · THE OFF-GRID HOME · 1 HOUR · EASY

A metal box blocks electromagnetic waves from reaching what's inside. A galvanized trash can is a cheap way to protect spare radios, a flashlight or a backup phone.

MATERIALS

- Galvanized steel trash can with a tight metal lid



- Cardboard to line the inside



- Aluminum tape



- Small electronics in plastic bags



1



Line It With Cardboard

Line the inside of the can and the lid with cardboard so no device touches metal.

2



Bag the Electronics

Put electronics in plastic bags, then inside the can.

3

Seal the Lid With Metal Tape

Close the lid and seal the seam all the way around with aluminum tape so metal meets metal.



4



Call the Phone Inside

Test: put a phone inside and call it. If it rings, the seal isn't tight. Fix gaps and try again.

5



Dry and Off the Floor

Store in a dry spot and leave it off the floor.

6



What to Keep Inside

Good things to keep inside: a hand-crank radio, two-way radios, a flashlight, a solar charge controller, a USB drive of family documents.

NOTES

A clean, sealed metal ammo can works the same way for smaller items.

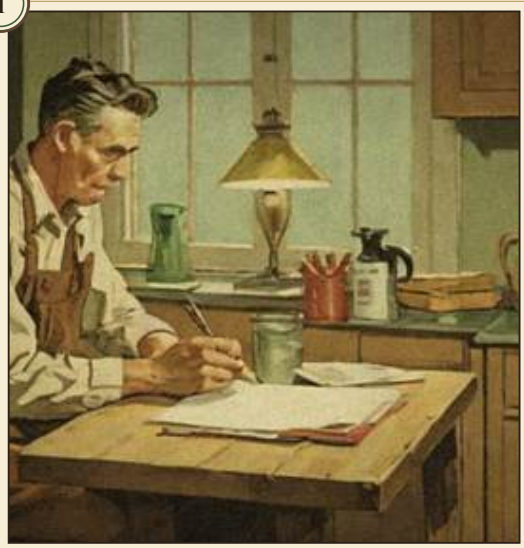
HOW TO

SIMPLE SOLAR BACKUP POWER

SKILL 34 · THE OFF-GRID HOME · 1 DAY · MEDIUM

A small solar setup can keep phones, lights, a radio and a CPAP running. Buy a ready-made 'power station' or build a basic kit.

1



Add Up the Watts

Add up what you need: watts of each device x hours per day. Phones and LED lights need little; fridges and heaters need a lot.

2



Battery First, Then Panel

Connect battery to charge controller first, then panel to controller. This order protects the controller.

3



Fuse Near the Positive

Place an inline fuse close to the battery positive terminal.

MATERIALS



100 to 200 watt solar panel



Charge controller (MPPT preferred)



A deep-cycle battery (LiFePO4 or AGM), 100 amp-hour



Small inverter (300 to 1000 watt)



Fuses, cable and connectors

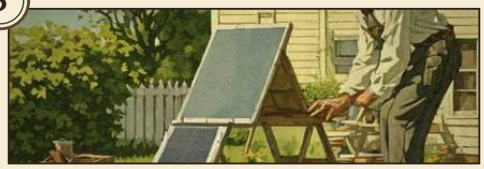
4



Short, Thick Cables to the Inverter

Connect the inverter directly to the battery with short, thick cables.

5



Face South, Tilt to Latitude

Aim the panel south (in the Northern Hemisphere) and tilt it to about your latitude.



6 ### *Run DC Straight From the Controller*

Run DC devices (USB, 12-volt) straight from the controller when possible; it wastes less power.

NOTES

IMPORTANT

Batteries can deliver dangerous current. Always fuse the circuit and never let positive and negative cables touch.

CHAPTER SIX

6

GARDEN AND HOMESTEAD

Growing food on any patch of ground







You don't need a farm to grow a good share of your food. A quarter acre, a backyard, or even a sunny porch can produce more than most people imagine when planned well.

IN THIS CHAPTER

35 Feeding a Family on a Quarter Acre
36 Building a Pallet Garden Fence
37 Dwarf Fruit Trees in Small Spaces
38 Composting the Easy Way
39 Saving Your Own Seeds
40 Keeping a Few Backyard Hens

HOW TO

FEEDING A FAMILY ON A QUARTER ACRE

SKILL 35 · GARDEN AND HOMESTEAD · A SEASON · MEDIUM

A quarter acre is about 10,900 square feet, roughly 100 x 109 feet. Laid out well it can grow vegetables, fruit, eggs and herbs for a family.

1



Map the Sun

Map the sun. Note where shade falls in morning, noon and late afternoon. Vegetables need 6 to 8 hours of sun.

2



Zone 1: By the Kitchen Door

Zone 1 (by the kitchen door): herbs, salad greens and tomatoes you pick daily.

3

Zone 2: The Main Beds

Zone 2: main vegetable beds, about 2,000 square feet. Grow what you eat most: potatoes, beans, squash, onions.



4



Zone 3: Fruit on the Edges

Zone 3: fruit trees and berry bushes along the edges and fences.

5



Zone 4: Hens, Compost, Rain

Zone 4: a small coop for 4 to 6 hens, a compost area and rain barrels.

6



Keep a Garden Record

Keep records each year of what grew well and what you ran short of.

NOTES

Potatoes, dry beans, winter squash and sweet potatoes give the most calories per square foot.

MATERIALS

- Graph paper



- A tape measure and stakes



- Seed catalogs



- A soil test kit



HOW TO

BUILDING A PALLET GARDEN FENCE

SKILL 36 · GARDEN AND HOMESTEAD · 1 DAY · EASY

Free wooden pallets can become a fence that keeps out dogs and rabbits and also grows food on its face.

MATERIALS



Heat-treated pallets (stamped 'HT', never 'MB')



Landscape fabric and a staple gun



Potting soil



Posts, screws, a drill



Seedlings: lettuce, herbs, strawberries

1



Check the Stamp: HT Only

Check the stamp. HT means heat-treated. MB (methyl bromide) pallets are chemically treated; don't use them for food.

2



Posts Every Four Feet

Set posts every 4 feet and screw the pallets to them standing upright.

3



Staple Fabric to the Back

Staple landscape fabric to the back and bottom of each pallet to make pockets between the slats.

4



Fill the Pockets With Soil

Fill the pockets with potting soil, packing it firmly.

5



Plant Between the Slats

Plant seedlings in the gaps between slats.



6 ### *Water From the Top*

Water gently from the top; it trickles down through all the rows.

NOTES

Shallow-rooted plants do best: lettuce, spinach, thyme, oregano, strawberries and nasturtiums.

HOW TO

DWARF FRUIT TREES IN SMALL SPACES

SKILL 37 · GARDEN AND HOMESTEAD · 1 HOUR PER TREE · EASY

Dwarf trees reach only 8 to 10 feet, start bearing in 2 to 3 years and can grow in a large pot on a patio.

1



Check Chill Hours and Pollination

Check the tree's 'chill hours' match your area, and whether it needs a second variety for pollination.

2



Twice as Wide, No Deeper

Dig a hole twice as wide as the root ball and no deeper.

3

Keep the Graft Above Soil

Keep the graft union (the bump on the trunk) 2 inches above the soil.



4



Backfill, Water, Mulch

Backfill with native soil mixed with compost, water well, and mulch in a ring away from the trunk.

5



Stake the First Year

Stake for the first year.

6



Thin the Fruit

Thin fruit to one per cluster in late spring so the tree doesn't overbear.

NOTES

A 'family' tree has several varieties grafted on one trunk, so it pollinates itself.

MATERIALS

- Dwarf or semi-dwarf tree (apple, pear, cherry, fig, citrus in warm areas)



- A 15 to 20 gallon pot with holes, or a sunny garden spot



- Compost and potting mix



- Stake and soft ties



- Mulch



HOW TO

COMPOSTING THE EASY WAY

SKILL 38 · GARDEN AND HOMESTEAD · ONGOING · EASY

Compost turns kitchen scraps and leaves into the best soil you'll ever have. Mix greens and browns, keep it damp, and nature does the rest.

MATERIALS



A bin: pallets wired together, or a simple pile 3 x 3 x 3 feet



Greens: scraps, grass clippings, coffee grounds



Browns: dry leaves, straw, shredded cardboard



A pitchfork and a hose

1



Coarse Browns at the Bottom

Start with a 6-inch layer of coarse browns at the bottom for air.

2



Greens and Browns in Layers

Add greens and browns in layers, about one bucket of greens to two of browns.

3



Damp Like a Wrung Sponge

Keep it damp like a wrung-out sponge.

4



Turn It With a Fork

Turn it with a fork every week or two to add air. A hot pile steams.

5



Dark, Crumbly, Earthy

Compost is ready in 2 to 6 months when dark, crumbly and earthy-smelling.



6 *Leave These Out*

Leave out meat, dairy, grease and pet waste; they attract pests.

NOTES

If it smells bad, add browns and turn it. If nothing happens, add greens and water.

HOW TO

SAVING YOUR OWN SEEDS

SKILL 39 · GARDEN AND HOMESTEAD · MINUTES AT HARVEST · EASY

Saving seed means you never need to buy them again, and your plants adapt to your land over the years.

1



Mark the Best Plant

Pick the best plant, not just the best fruit, and mark it with a ribbon.

2



Beans: Dry on the Vine

Beans And Peas: let pods dry brown on the vine, then shell.

3

Tomatoes: Ferment and Rinse

Tomatoes: scoop seeds into a jar with a little water, let ferment 3 days, rinse, and dry on a plate.



4



Squash and Peppers: Scoop and Dry

Squash And Peppers: scoop from fully ripe fruit, rinse and dry.

5



Lettuce: Shake Into a Bag

Lettuce And Herbs: let them flower and go to seed, then shake heads into a paper bag.

6



Label, Jar, Keep Cool

Dry seeds a week, then store in labeled envelopes inside a sealed jar in a cool, dark place.

NOTES

Add a packet of dried milk powder or silica gel to the jar to absorb moisture.

MATERIALS

- Open-pollinated or heirloom plants (not hybrids marked F1)



- Paper envelopes and labels



- Glass jars



- A dry, cool place



HOW TO

KEEPING A FEW BACKYARD HENS

SKILL 40 · GARDEN AND HOMESTEAD · 1 WEEKEND SETUP · MEDIUM

Four hens give a family of four roughly two dozen eggs a week, eat kitchen scraps and make compost gold.

MATERIALS


A coop with 4 square feet per bird inside

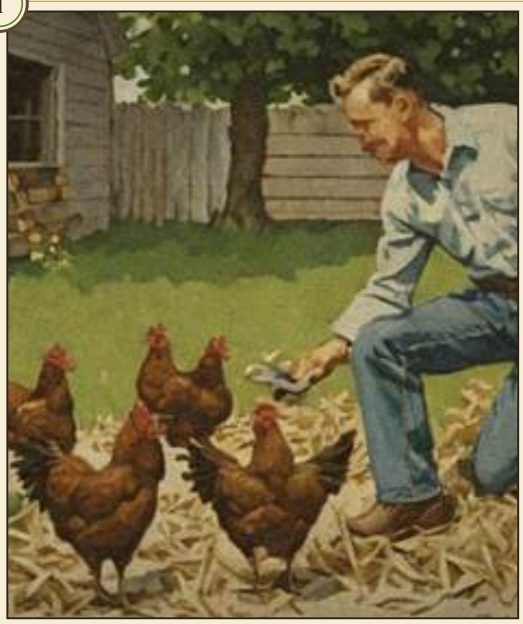

A run with 10 square feet per bird, covered with hardware cloth


Roosting bar, nest boxes (1 per 4 hens)


Feeder, waterer, layer feed, oyster shell


Check local laws on backyard poultry

1



Three to Six Hens

Start with 3 to 6 hens; chickens are social and should never be kept alone.

2



Hardware Cloth, Buried

Build the run with buried hardware cloth (not chicken wire) to stop predators digging in.

3



Feed, Water, Calcium

Give them layer feed, fresh water daily, and crushed shells or oyster shell for calcium.

4



Lock Up at Night

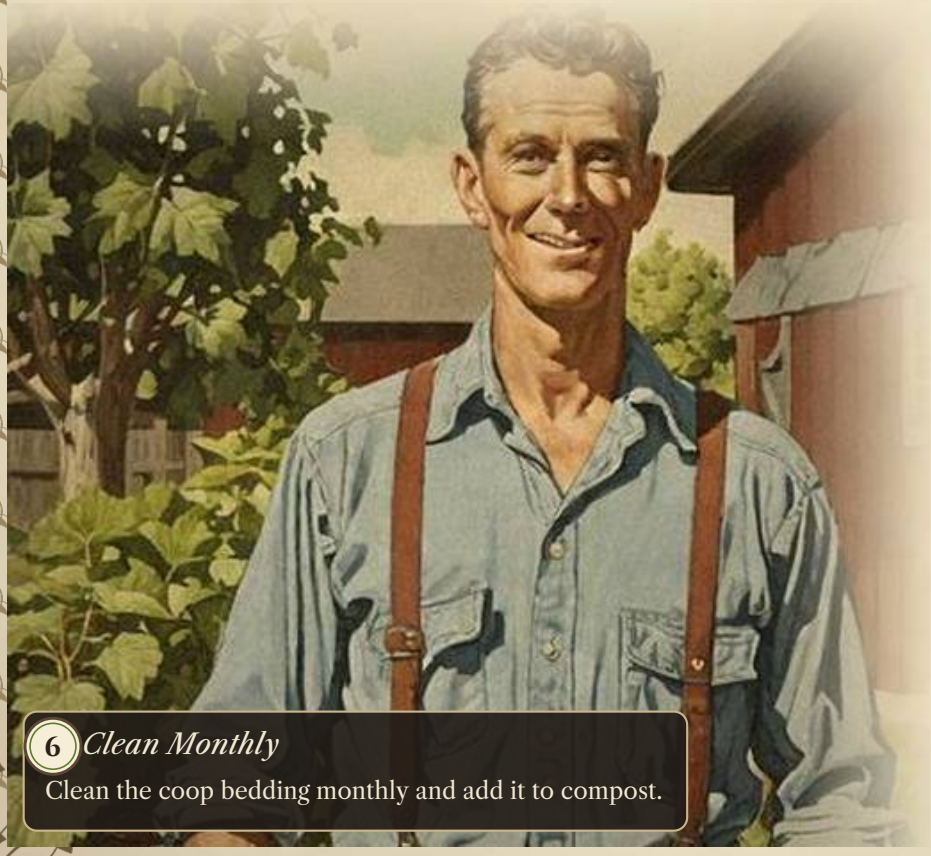
Lock them in the coop every night.

5



Collect Eggs Daily

Collect eggs daily. Don't wash them until just before use if you want to store them.



6 ### *Clean Monthly*

Clean the coop bedding monthly and add it to compost.

NOTES

Pine shavings or straw on the floor and a 'deep litter' method keeps the coop clean and warm in winter.

CHAPTER SEVEN

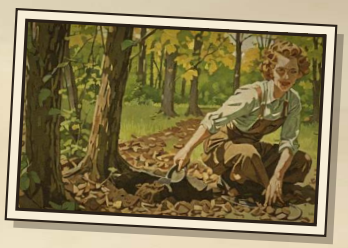
7

OUTDOORS AND WILDERNESS

Fire, shelter and finding your way







A walk in the woods can turn into a night out because of a twisted ankle or a wrong turn. These skills let you stay warm, get found, and enjoy the outdoors with confidence.

IN THIS CHAPTER

41 Building and Lighting a Campfire
42 Fire with a Ferro Rod
43 The Dakota Fire Hole
44 The Debris Hut
45 Five Knots Worth Knowing
46 Finding Direction Without a Compass
47 Signaling for Rescue

HOW TO

BUILDING AND LIGHTING A CAMPFIRE

SKILL 41 · OUTDOORS AND WILDERNESS · 20 MINUTES · EASY

Fire needs three things: fuel, air and heat. Most failed fires are short on small, dry fuel.

MATERIALS

- Tinder: dry grass, birch bark, cotton balls with petroleum jelly



- Kindling: pencil-lead to thumb thick



- Fuel wood: wrist thick and up



- Matches, a lighter or a ferro rod



- Water and a shovel



1



Clear a Safe Spot

Pick a cleared spot away from overhanging branches. Use a fire ring or clear to bare soil.

2



Gather Everything First

Gather everything BEFORE striking a match: two big handfuls of tinder, two armloads of kindling.

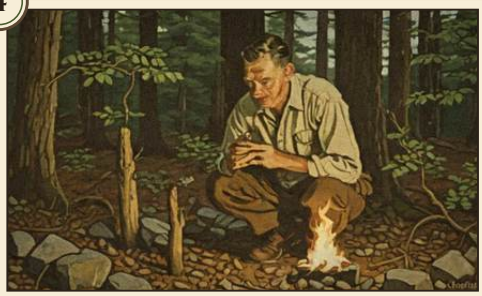
3

Tinder Ball, Kindling Teepee

Make a loose ball of tinder. Lean kindling over it in a teepee, leaving a door on the windward side.



4



Light From the Bottom

Light the tinder from the bottom. Blow gently at the base.

5



Bigger Sticks Only When Ready

Add bigger sticks only once the small ones burn well.

6



Drown, Stir, Drown Again

To put it out: drown, stir, and drown again until it's cold to the touch.

NOTES

Dead branches still on a tree ('standing deadwood') are drier than anything on the ground.

HOW TO

FIRE WITH A FERRO ROD

SKILL 42 · OUTDOORS AND WILDERNESS · 10 MINUTES TO LEARN · EASY

A ferrocerium rod throws 5,000 F sparks, works wet, and lasts thousands of strikes. It's the most reliable fire starter you can carry.

1



Fluff a Tinder Nest

Fluff the tinder into a loose nest on dry ground or bark.

2



Rod Tip in the Tinder

Hold the rod tip right in the tinder, at a 45 degree angle.

3



Pull the Rod Back

Press the striker on the rod and PULL THE ROD back, rather than pushing the striker forward. This keeps the tinder from scattering.

MATERIALS

- Ferro rod and striker (or the back of a knife)



- Very fine tinder: cotton ball, cattail fluff, fatwood shavings



4



A Shower of Sparks

Scrape firmly. A shower of sparks lands directly on the tinder.

5

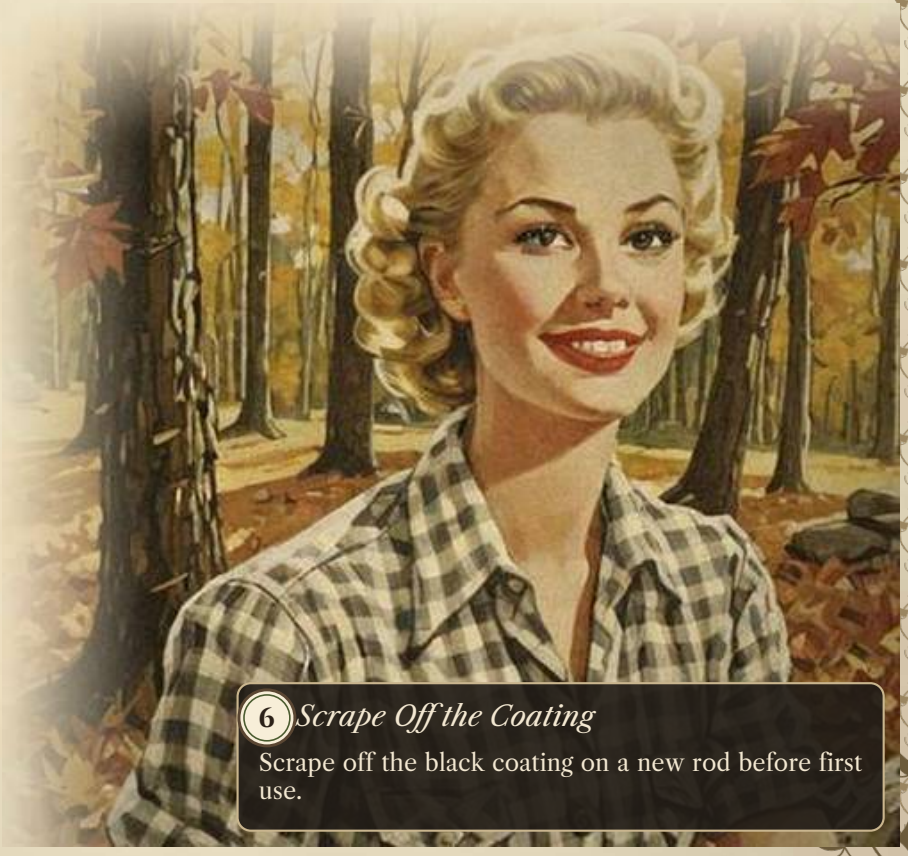


Cup and Blow Gently

When it catches, cup your hands and blow gently, then lay it under your kindling.

NOTES

Fatwood (resin-soaked pine stump) catches even when damp. Shave fine curls with your knife.



6 *Scrape Off the Coating*

Scrape off the black coating on a new rod before first use.

HOW TO

THE DAKOTA FIRE HOLE

SKILL 43 · OUTDOORS AND WILDERNESS · 30 MINUTES · MEDIUM

Two holes connected by a tunnel make a fire that burns hot, uses little wood, hides its flame and resists wind. Plains peoples used this trick for centuries.

MATERIALS

- A small shovel or trowel



- Firm soil (not sand or rocky ground)



- Small dry fuel



1



Dig the Fire Hole

Dig a fire hole about 1 foot wide and 1 foot deep. Save the sod.

2



A Smaller Air Hole Upwind

About a foot away, on the side the wind blows from, dig a smaller air hole about 6 inches wide.

3

Tunnel Between Them

Tunnel between the bottoms of the two holes with a stick or your hand.



4



Light It at the Bottom

Light the fire at the bottom of the fire hole. Air rushes in through the tunnel.

5



Green Sticks for a Pot

Lay green sticks across the top to rest a pot.

6



Leave No Trace

When done, put it out with water, fill the holes and replace the sod. You'll leave no trace.

NOTES

The flame stays below ground level, making it hard to see from a distance and safer in wind.

HOW TO

THE DEBRIS HUT

SKILL 44 · OUTDOORS AND WILDERNESS · 2 TO 3 HOURS · MEDIUM

Lost without a tent? A debris hut made of sticks and leaves can keep you alive through a cold night. Think of it as a sleeping bag made of forest.

1



Prop the Ridgepole

Prop one end of the ridgepole about waist height on a stump or forked stick. The other end rests on the ground.

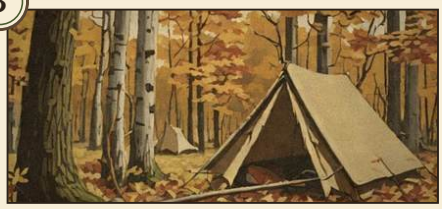
2



Check the Fit

Lie under it to check fit: it should be just bigger than your body. Smaller is warmer.

3



Lean the Ribs

Lean sticks along both sides close together to make ribs.

MATERIALS



A sturdy ridge pole longer than you are tall



A stump, rock or forked stick to prop it



Lots of sticks



Huge amounts of dry leaves, pine needles or grass

4



Pile on the Leaves

Pile leaves over the frame at least 2 feet thick; 3 feet is better. Then add more.

5

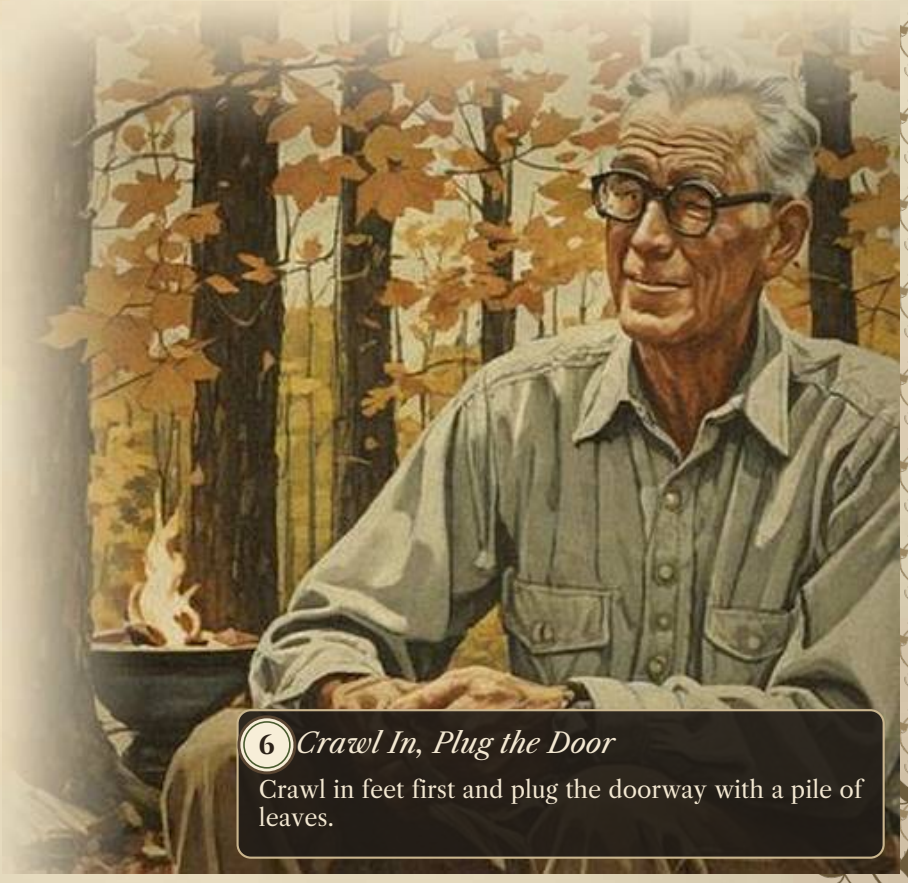


Stuff the Inside

Stuff the inside with a thick, fluffy bed of dry leaves. The ground steals more heat than the air.

NOTES

If you're lost, stay put after building your shelter. Searchers find people who stay in one place much sooner.



6 ### *Crawl In, Plug the Door*

Crawl in feet first and plug the doorway with a pile of leaves.

HOW TO

FIVE KNOTS WORTH KNOWING

SKILL 45 · OUTDOORS AND WILDERNESS · 1 EVENING · EASY

MATERIALS

- A 6-foot length of rope or paracord



You can do nearly any job with five knots. Practice them with a short rope until you can tie them with your eyes closed.

1



Square Knot

Right over left, left over right. Joins two ends of the same size, like tying a bundle.

2



Bowline

Makes a fixed loop that won't slip or jam. 'The rabbit comes out of the hole, around the tree, and back down the hole.'

3

Clove Hitch

Two loops around a post, the second crossing over the first. Quick to tie a rope to a pole.



4



Taut-Line Hitch

An adjustable loop for tent lines and clotheslines; slides to tighten, holds under load.

5



Trucker's Hitch

A loop pulley that lets you cinch a load down tight.

6



Finish With a Half Hitch

Finish important knots with a half hitch to keep them from shaking loose.

NOTES

Paracord can be pulled apart into 7 inner strands for fishing line, sewing thread or snares.

HOW TO

FINDING DIRECTION WITHOUT A COMPASS

SKILL 46 · OUTDOORS AND WILDERNESS · 15 MINUTES · EASY

The sun and stars have guided travelers forever. These methods give you a rough direction good enough to walk out.

1



Shadow Stick: First Stone

Shadow Stick: push the stick upright into flat ground. Mark the tip of its shadow with a stone.

2



Wait 15 Minutes

Wait 15 to 20 minutes and mark the new shadow tip with a second stone.

3



West to East

A line from the first stone to the second runs roughly west to east. Stand with the first on your left: you're facing north.

MATERIALS



A straight stick about 3 feet long



Two small stones



A clear sky

4



The North Star

North Star: find the Big Dipper. The two stars at the outer edge of its cup point to Polaris, the North Star, about 5 cup-widths away.

5



The Watch Method

WATCH METHOD (northern hemisphere): point the hour hand at the sun. Halfway between it and 12 is south.



6 ### *Walk Landmark to Landmark*

Pick a landmark in your direction and walk to it, then pick the next.

NOTES

Moss grows on all sides of trees. Don't trust it for direction.

HOW TO

SIGNALING FOR RESCUE

SKILL 47 · OUTDOORS AND WILDERNESS · MINUTES · EASY

The faster rescuers see you, the sooner you're home. Signals in threes mean 'help' all over the world.

MATERIALS

- A whistle



- A signal mirror or any shiny object



- Bright clothing or a tarp



- Fire and green branches for smoke



1



Whistle: Three Blasts

Whistle: three sharp blasts, pause, repeat. It carries much farther than shouting and saves your voice.

2



Mirror: Aim Through the V

Mirror: hold your other hand out in a V toward a plane or distant hill. Tilt the mirror until the sunlight spot falls between your fingers, then wiggle it.

3

A Giant SOS on the Ground

Ground: lay out a huge 'SOS' or 'X' with rocks, logs or clothes in an open area. Make letters at least 10 feet tall.



4



Smoke by Day

Smoke: green leaves on a hot fire make white smoke that shows against dark forest.

5



Three Fires in a Triangle

Three fires in a triangle are an international distress signal.

6



Three Flashes at Night

At night, flash a light three times, pause, repeat.

NOTES

Always leave a trip plan with someone: where you're going, your route and when you'll be back.

CHAPTER EIGHT

8

SAFETY, MIND AND COMMUNITY

Staying calm, aware and connected







Independence isn't only about gear. Keeping a clear head, paying attention, handling difficult people and knowing your neighbors matter just as much when things get hard.

IN THIS CHAPTER

48 Escaping a Sinking or Trapped Car
49 Spotting Hidden Cameras in a Rental
50 Situational Awareness
51 Handling Insults and Difficult People
52 Building Calm Confidence
53 Your Family Emergency Plan
54 Radio When the Phones Go Down

HOW TO

ESCAPING A SINKING OR TRAPPED CAR

SKILL 48 · SAFETY, MIND AND COMMUNITY · SECONDS · EASY

If a car goes into water, you may have under a minute before it sinks. Don't wait for it to fill. Remember: Seatbelt, Window, Out.

MATERIALS

- A spring-loaded window punch with a belt cutter, mounted within reach



- Glow-in-the-dark trunk release (built into most cars since 2002)



1



Seatbelt First

Seatbelt: unbuckle at once. Cut it if stuck.

2



Window: Open or Break It

Window: open the side window immediately while power works. If not, break it with the punch at a lower corner. Windshields are laminated and won't break easily.

3



Oldest Child First

Children: unbuckle the oldest first so they can help, then the youngest. Push them out ahead of you.

4



Out the Window, Up

OUT: climb out through the window and swim to the surface. Don't try the door; water pressure holds it shut.

5



Fire: Get Out and Away

Fire: get everyone out and at least 100 feet away. Don't go back for belongings.



6 *Trunk: Find the Glowing Handle*

Trunk: if locked in a trunk, find the glowing T-handle and pull. In older cars, kick out a tail-light and wave through the hole.

NOTES

Practice the motions with your family in the driveway. Rehearsal beats panic.

HOW TO

SPOTTING HIDDEN CAMERAS IN A RENTAL

SKILL 49 · SAFETY, MIND AND COMMUNITY · 15 MINUTES · EASY

Most rental hosts are honest, but hidden cameras do turn up. A quick sweep protects your privacy wherever you stay.

1



Check the Usual Spots

Check the likely spots: smoke detectors, clocks, chargers, picture frames, air vents and anything facing the bed or shower.

2



Lights Off, Flashlight On

Turn off the room lights. Shine a flashlight slowly around the room; a camera lens reflects a small bright dot.

3

Look Through the Phone Camera

Look through your phone camera in the dark. Many hidden cameras use infrared light, which some phone cameras show as purple dots (front cameras often work best).



4



Scan the Wi-Fi List

Look at the list of Wi-Fi devices on the network or nearby for camera names.

5



Unplug or Cover It

Unplug or cover anything suspicious with a towel.

6



Photograph, Leave, Report

If you find a camera in a private area, take photos, leave, and report it to the platform and police.

NOTES

Many rental sites ban indoor cameras outright and require hosts to disclose any outdoor ones.

MATERIALS

- A flashlight or phone light



- Your phone's camera



- A Wi-Fi network list



HOW TO

SITUATIONAL AWARENESS

SKILL 50 · SAFETY, MIND AND COMMUNITY · A HABIT · EASY

Most trouble can be avoided by noticing it early. Awareness isn't fear; it's simply paying attention.

MATERIALS

- Nothing but practice



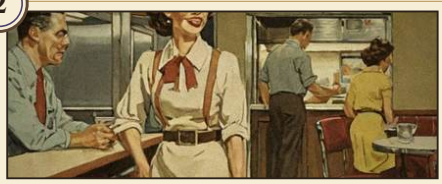
1



Eyes Up, Phone Down

Keep your eyes up and phone down in parking lots, transit and crowds.

2



Note Two Ways Out

When you enter a place, note two ways out.

3



Learn What Normal Looks Like

Learn what normal looks like in a place. Then notice what's out of place: someone watching instead of doing, a person who mirrors your moves, a car circling.

4



Trust the Gut, Act Early

Trust the gut feeling that something is wrong. Act on it early: cross the street, enter a shop, change your route.

5



Keys in Hand, Walk With Purpose

Walk with purpose and keep keys in hand near your car or door.



6 *Play "What If"*

Play 'what if' games in your head: what would I do if a fire alarm went off right now?

NOTES

Security pros use color codes: White (unaware), Yellow (relaxed but alert), Orange (something's off), Red (act). Live in Yellow.

HOW TO

HANDLING INSULTS AND DIFFICULT PEOPLE

SKILL 51 · SAFETY, MIND AND COMMUNITY · A HABIT · MEDIUM

Stress brings out the worst in people. Staying calm keeps you in control and often defuses the other person too.

1



Pause and Breathe

Pause and breathe before replying. A slow breath out calms your body.

2



Don't Take the Bait

Don't take the bait. A calm 'Maybe so' or 'I see it differently' gives an insult nothing to grip.

3

Name What Is Happening

Name what's happening without heat: 'It sounds like you're really frustrated.'



4



Set a Clear Limit

'I'm happy to talk, but not if you're shouting.' Then follow through.

5



Keep It Short and in Writing

With someone who always needs to win or blames others, keep conversations short, factual and in writing when possible.

6



Walk Away

Walk away when nothing good can come of it. Leaving is a strength, not a defeat.

NOTES

Think of a 'gray rock': give dull, brief answers to people who feed on drama, and they lose interest.

MATERIALS

- Patience



- A few practiced phrases



HOW TO

BUILDING CALM CONFIDENCE

SKILL 52 · SAFETY, MIND AND COMMUNITY · DAILY · EASY

Confidence comes from competence. Every skill in this book that you actually practice adds to it.

MATERIALS

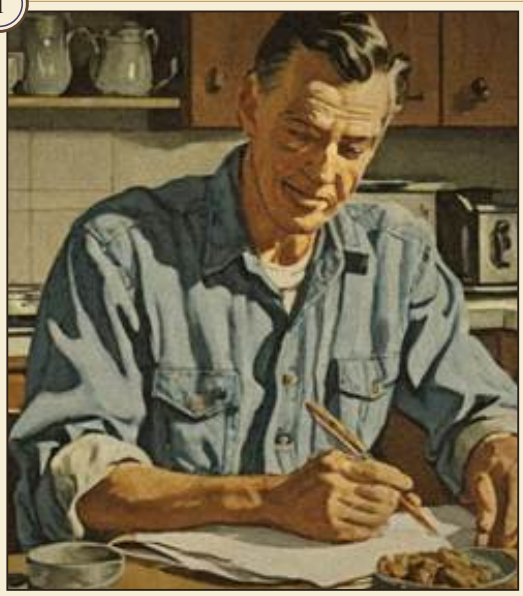
- A notebook



- A small goal each week



1



One Skill a Week

Pick one skill from this book each week and actually do it, start to finish.
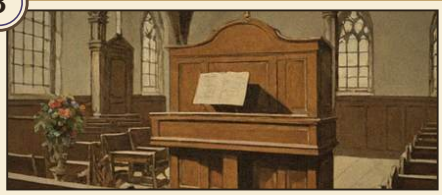
2



Write Down What Worked

Write down what worked and what you'd change. Seeing progress builds belief.

3



Stand Tall, Speak Slowly

Stand tall, slow your speech and make eye contact; your mind follows your body.

4



Rehearse What You Can

Prepare. Most nerves come from the unknown; rehearse what you can.

5



Get Uncomfortable on Purpose

Get a little uncomfortable on purpose: cold showers, a long walk, a new conversation.



6 *Teach Someone Else*

Teach someone else. Explaining a skill locks it in for you.

NOTES

Box breathing for stress: breathe in 4 seconds, hold 4, out 4, hold 4. Repeat four times.

HOW TO

YOUR FAMILY EMERGENCY PLAN

SKILL 53 · SAFETY, MIND AND COMMUNITY · 1 EVENING · EASY

In a disaster, phones fail and families get split up. A plan made at the kitchen table now prevents panic later.

1



Two Meeting Places

Choose two meeting places: one right outside your home, one outside your neighborhood.

2



An Out-of-State Contact

Pick an out-of-state contact everyone calls or texts; long-distance lines often work when local ones are jammed.

3

Numbers on a Card

Write down key numbers on a card for every wallet and backpack. Don't count on phone memory.



4



Two Routes Out of Town

Plan routes out of town in two different directions and mark them on the paper map.

5



Know the School's Plan

Know your children's school plans and who's allowed to pick them up.

6



Drill Twice a Year

Practice: run a fire drill and a 'meet at the second spot' drill twice a year.

NOTES

Text messages often get through when calls don't. Agree on short code words like 'SAFE HOME'.

MATERIALS

- Paper and a pen



- Copies of IDs and important papers



- A paper map of your area



HOW TO

RADIO WHEN THE PHONES GO DOWN

SKILL 54 · SAFETY, MIND AND COMMUNITY · A FEW WEEKENDS · MEDIUM

When cell towers fail, radio still works. Even a cheap weather radio keeps you informed, and licensed amateur ('ham') radio lets you talk across the country.

MATERIALS



Hand-crank/solar weather radio with NOAA channels



FRS two-way radios (no license needed)



Optional: GMRS radios (simple license) or ham radio (test required)

1



A Weather Radio on the Counter

Get a NOAA weather radio with alerts and keep it on the kitchen counter.

2



Family Walkie-Talkies

Buy a set of FRS walkie-talkies for family use around the neighborhood. Agree on a channel.

3



GMRS for Longer Range

For longer range, get a GMRS license (no test, covers your whole family) and use repeaters.

4



Study for the Ham License

To talk worldwide, study for the amateur Technician license; it's a 35-question test.

5



Join a Local Club

Join a local radio club. Members help newcomers and run emergency nets.



6 ### *Spares in the Faraday Can*

Store radios and spare batteries in your Faraday can.

NOTES

Write your family's channel and check-in times (say, top of every hour) on the emergency card.

CHAPTER NINE

9

FIX-IT SKILLS

Mending, sharpening and repairing instead of replacing







'Use it up, wear it out, make it do, or do without.' That old saying kept homes running through hard times. A few simple repairs save money and keep good things out of the landfill.

IN THIS CHAPTER

55 Fixing a Stuck or Broken Zipper
56 Sharpening a Knife on a Stone
57 Sewing a Button and Mending a Tear
58 Fixing a Dripping Faucet
59 The Essential Home Tool Kit
60 Darning Socks and Patching Jeans

HOW TO

FIXING A STUCK OR BROKEN ZIPPER

SKILL 55 · FIX-IT SKILLS · 10 MINUTES · EASY

Most 'broken' zippers just need cleaning or a squeeze. Don't throw out a good coat over a zipper.

MATERIALS

- Graphite pencil, bar soap or candle wax



- Small pliers



- Tweezers



- Replacement slider (optional)



1



Free the Caught Fabric

Stuck: pull fabric out of the teeth with tweezers. Don't yank.

2



Graphite, Soap or Wax

Rub graphite, soap or wax along the teeth on both sides of the slider. Work it gently back and forth.

Squeeze a Spread Slider

SPLITTING (teeth open behind slider): the slider has spread. Squeeze each side gently with pliers, a little at a time, and test.

3



4



A New Pull

Missing Pull: thread a paper clip, key ring or loop of cord through the hole.

5



Slider Off? Notch and Rethread

Slider Off One Side: cut a notch in the teeth near the bottom, slide it back on both sides, and sew a new stop.

6



Close Them Before Washing

Wash and close zippers before laundering to keep teeth straight.

NOTES

A drop of clear nail polish on the top stitches keeps a zipper on jeans from creeping down.

HOW TO

SHARPENING A KNIFE ON A STONE

SKILL 56 · FIX-IT SKILLS · 20 MINUTES · MEDIUM

A sharp knife is safer than a dull one because it doesn't slip. A good stone and a steady angle are all you need.

1



Soak the Stone

Soak a water stone for 10 minutes. Set it on a damp towel so it won't slide.

2



Color the Edge

Color the edge with marker. Grind marks will show if your angle is right.

3



Hold About 20 Degrees

Hold the blade at about 20 degrees (two stacked coins under the spine is close).

MATERIALS



Combination whetstone (about 1000/6000 grit)



Water or honing oil (whichever the stone is made for)



A damp towel to hold the stone



A marker

4



Slice a Thin Layer Off the Stone

Push the edge across the coarse side as if slicing a thin layer off the stone, heel to tip. Do 10 strokes, then 10 on the other side.

5



Feel for the Burr

Feel for a tiny burr along the opposite edge. When it runs the full length, switch to the fine side and repeat with lighter strokes.

NOTES

Test on a sheet of paper: a sharp edge slices cleanly without tearing.



6 ### *Strop on Leather*

Finish with a few strokes on a leather belt or newspaper to remove the burr.

HOW TO

SEWING A BUTTON AND MENDING A TEAR

SKILL 57 · FIX-IT SKILLS · 15 MINUTES · EASY

Two stitches cover most mending: a button stitch and a ladder stitch for split seams that's almost invisible.

MATERIALS

- Needle and thread matched to the fabric



- Scissors



- A pin or toothpick



- Buttons



1



Thread, Double, Knot

Thread the needle, double the thread and knot the end.

2



Sew Over a Toothpick

Button: come up from the back where the button goes. Lay a toothpick across the button and sew over it 6 to 8 times through the holes.

Wind a Shank, Knot Behind

Remove the toothpick, wind the thread around the stitches under the button to make a 'shank', and knot on the back.

3



4



Ladder Stitch a Seam

LADDER STITCH for a seam: turn edges under, then take a tiny stitch straight across inside one fold, then the other, like rungs of a ladder.

5



Pull Gently and It Disappears

Pull the thread gently every few stitches; the seam closes and the stitches disappear.

6



Patch a Hole From Behind

Patch: for holes, sew a scrap behind the hole with small running stitches all round.

NOTES

Keep a sewing kit with a few needles, safety pins and black, white and khaki thread in every bag.

HOW TO

FIXING A DRIPPING FAUCET

SKILL 58 · FIX-IT SKILLS · 1 HOUR · MEDIUM

A faucet that drips once a second wastes over 3,000 gallons a year. Most drips are a worn washer or cartridge, a cheap part.

1



Shut Off the Water

Shut off water at the valves under the sink. Open the faucet to drain it.

2



A Rag in the Drain

Put a rag in the drain so small parts can't fall in.

3



Pry the Cap, Lift the Handle

Pry off the decorative cap on the handle and remove the screw beneath. Lift off the handle.

MATERIALS



Adjustable wrench, screwdrivers



Replacement washers, O-rings or cartridge for your faucet brand



Plumber's grease



A rag and a towel

4



Pull the Stem, Keep Parts in Order

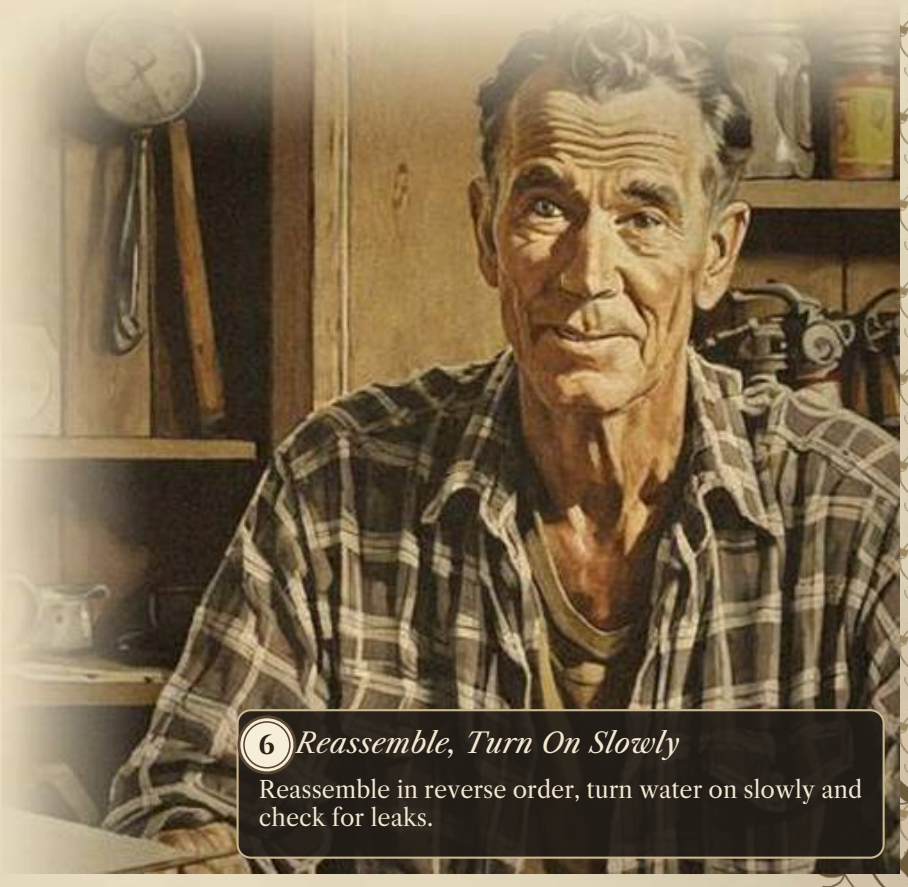
Unscrew the packing nut or retaining clip. Pull out the stem or cartridge, keeping parts in order on the towel.

5



Match the Part at the Store

Take the old washer or cartridge to the hardware store to match it. Replace and coat O-rings with plumber's grease.



6 ### *Reassemble, Turn On Slowly*

Reassemble in reverse order, turn water on slowly and check for leaks.

NOTES

Take a phone photo at each step so reassembly is easy.

HOW TO

THE ESSENTIAL HOME TOOL KIT

SKILL 59 · FIX-IT SKILLS · 1 SHOPPING TRIP · EASY

With about fifteen tools you can fix most things in a house. Buy good ones once; they'll last a lifetime.

MATERIALS

- A sturdy toolbox



- The list below



1



Hand Tools

Claw hammer, screwdriver set (flat and Phillips), adjustable wrench, slip-joint and needle-nose pliers, utility knife, tape measure, level, hand saw.

2



A Cordless Drill

Power: a cordless drill/driver with a bit set.

3

Fastening

Assorted screws, nails, wall anchors, zip ties, duct tape, wire.



4



Plumbing and Electric

Plunger, plumber's tape, a voltage tester, spare fuses.

5



Safety

Work gloves, safety glasses, dust masks, a flashlight.

6



A Notebook in the Box

Keep a small notebook in the box listing paint colors, filter sizes and bulb types for your house.

NOTES

Turn off the breaker AND test with a voltage tester before touching any wiring.

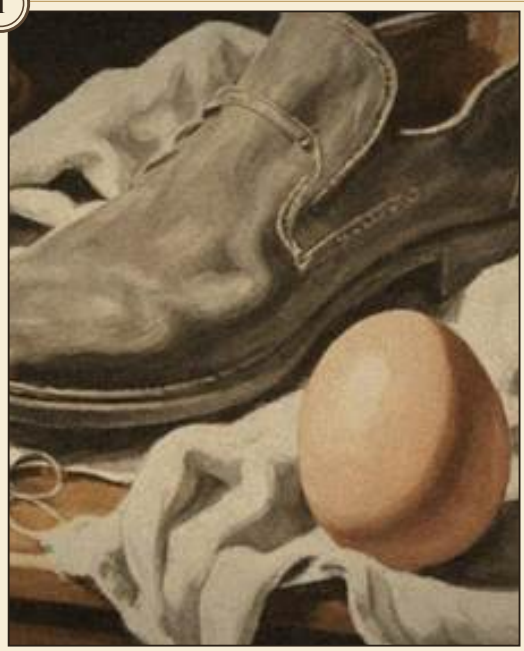
HOW TO

DARNING SOCKS AND PATCHING JEANS

SKILL 60 · FIX-IT SKILLS · 30 MINUTES · EASY

Darning weaves new threads across a hole. It's relaxing work and makes favorite wool socks last for years.

1



Egg Inside the Sock

Slip the egg inside the sock under the hole.

2



Stitches Up and Down

Run stitches up and down across the hole, starting a half-inch beyond it on solid fabric. Leave tiny loops at the ends so it can stretch.

3



Weave Across

Turn and weave across those threads at right angles: over one, under one, then back the other way.

MATERIALS



Darning needle and yarn close to the sock's weight



A darning egg, light bulb or smooth stone



Iron-on or sew-on patches for jeans

4



Pack the Rows Close

Pack the rows close so the patch is firm but flat.

5



Jeans: Patch Behind the Hole

Jeans: put the patch inside behind the hole, then stitch back and forth across the worn area by hand or machine.

NOTES

Mend early. A thin spot is easy to darn; a big hole takes much longer.



6 *Sashiko: Make It Handsome*

Japanese 'sashiko' running stitches in contrasting thread make repairs look intentional and handsome.

APPENDIX

Charts, checklists and quick references

A. WATER DISINFECTION WITH BLEACH

Use plain, unscented household bleach. Check the label for strength. Double the amount for cloudy water, stir, and wait 30 minutes. Water should have a slight chlorine smell.

WATER	6% BLEACH	8.25% BLEACH
1 quart / liter	2 drops	2 drops
1 gallon	8 drops	6 drops
2 gallons	16 drops (1/4 tsp)	12 drops
5 gallons	40 drops (1/2 tsp)	32 drops
55 gallons	about 1/3 cup	about 1/4 cup

How much water to store: 1 gallon per person per day for drinking and basic hygiene. Add 1/2 gallon per day for a medium dog. Hot climates, nursing mothers and sick people need more.

B. PANTRY SHELF-LIFE CHART

*Approximate times for best quality when stored cool (under 70 F), dry and dark. Rotate: first in, first out.

FOOD	UNOPENED SHELF LIFE*	STORE IT
White rice	25 to 30 years	Sealed, oxygen absorbers
Dry beans	10 to 30 years	Sealed buckets or #10 cans
Rolled oats	25 to 30 years	Sealed, oxygen absorbers
Wheat berries	25+ years	Sealed buckets
White flour	1 to 2 years	Cool, sealed; freeze 3 days first
Honey	Indefinite	Sealed jar; may crystallize
Salt	Indefinite	Dry, sealed
White sugar	Indefinite	Dry, sealed
Pressure-canned meat	1 to 5 years	Cool, dark shelf
Home-canned fruit	1 to 2 years (best quality)	Cool, dark shelf
Powdered milk	2 to 20 years (by type)	Sealed, cool
Pemmican	1 to 5+ years	Cool, dark, wrapped
Hardtack	Many years	Airtight tin
Water-glassed eggs	8 to 12 months	Under 70 F
Lard (rendered)	6 to 12 months	Cool pantry; longer chilled
Vegetable oil	1 to 2 years	Dark, cool
Coffee (whole bean, sealed)	1 to 2 years	Dark, cool

C. CHECKLISTS

Copy these pages and tick items off as you build your kits. Check expiry dates and refresh water twice a year, when the clocks change.

72-HOUR GO-BAG (PER PERSON)

- ☐ 3 gallons water, or 1 liter bottle plus a filter
- ☐ 3 days of no-cook food
- ☐ Change of clothes, rain jacket, warm hat
- ☐ Emergency blanket and poncho
- ☐ Headlamp with spare batteries
- ☐ First-aid kit and 7 days of medicines
- ☐ Copies of ID, insurance and contacts in a zip bag
- ☐ Cash in small bills
- ☐ Phone charger and battery bank
- ☐ Multi-tool or knife
- ☐ Whistle, lighter, ferro rod
- ☐ Toothbrush, soap, toilet paper, masks
- ☐ Paper map and family plan card

BUG-IN HOME KIT (2 WEEKS)

- ☐ 1 gallon water per person per day
- ☐ Two weeks of pantry food you normally eat
- ☐ Manual can opener
- ☐ Camp stove with fuel (outdoor use only)
- ☐ Battery/crank weather radio
- ☐ Lanterns, headlamps, batteries
- ☐ Carbon monoxide and smoke alarms with fresh batteries
- ☐ Fire extinguisher (ABC)
- ☐ Bleach, trash bags, buckets for sanitation
- ☐ Warm bedding for every person
- ☐ Basic tool kit and duct tape
- ☐ Plastic sheeting
- ☐ Pet food and supplies

CAR KIT

- ☐ Window punch and belt cutter
- ☐ Jumper cables or jump pack
- ☐ Tire inflator and plug kit
- ☐ Water and snack bars
- ☐ Blanket, gloves, hat
- ☐ Flashlight
- ☐ First-aid kit
- ☐ Ice scraper and small shovel
- ☐ Reflective triangles
- ☐ Phone charger

D. COMMON WILD EDIBLES

Never eat any wild plant unless you are 100% sure what it is. Learn from a local expert and a regional field guide, avoid roadsides and sprayed areas, and try a small amount first.

PLANT	EDIBLE PART	SEASON	LOOK-ALIKE WARNING
Dandelion	Leaves, flowers, roots	Spring to fall	Few; check for spraying
Broadleaf plantain	Young leaves, seeds	Spring to fall	Few
Chickweed	Leaves and stems	Cool months	Scarlet pimpernel (red flowers)
Purslane	Leaves and stems	Summer	Spurge (milky sap)
Wild violet	Leaves and flowers	Spring	Avoid any with yellow-only flowers in doubt
Clover	Flowers, young leaves	Spring to summer	Few; eat in moderation
Lamb's quarters	Leaves	Summer	Nightshades (avoid if unsure)
Cattail	Shoots, pollen, root	Spring to fall	Iris (no brown seed head)
Blackberry/raspberry	Berries	Summer	Few true look-alikes
Wild garlic/onion	Bulbs and leaves	Spring	Death camas: NO onion smell
Acorns (oak)	Nut, after leaching	Fall	Must leach tannins in water
Black walnut	Nut	Fall	Husk stains heavily

E. KITCHEN AND HOMESTEAD CONVERSIONS

MEASURE	EQUALS
3 teaspoons	1 tablespoon
16 tablespoons	1 cup
2 cups	1 pint
2 pints	1 quart (about 0.95 liter)
4 quarts	1 gallon (about 3.8 liters)
1 gallon of water	about 8.3 pounds
55-gallon drum of water	about 460 pounds
1 ounce	about 28 grams
1 pound	about 454 grams
1 inch	2.54 centimeters
1 foot	about 30.5 centimeters
1 acre	43,560 square feet
1/4 acre	10,890 square feet
Water boils (sea level)	212 F / 100 C
Water freezes	32 F / 0 C
F to C	subtract 32, then x 5/9
C to F	x 9/5, then add 32

F. FAMILY EMERGENCY CARD

Fill this in together, then copy it for every wallet, backpack and glove box.

Family name	
Home address	
Meeting place #1 (near home)	
Meeting place #2 (outside neighborhood)	
Out-of-state contact (name and phone)	
Doctor / clinic	
Pharmacy	
Allergies and medicines	
School / daycare plan	
Family radio channel and check-in time	
Code word for "we're safe"	
Pet information	
Insurance policy numbers	
Other notes	

IMPORTANT NUMBERS

Emergency	911
Poison Control (US)	1-800-222-1222
Police (non-emergency)	
Fire department	
Power company outage line	
Water / gas utility	
Neighbor	
Neighbor	

G. GLOSSARY

Botulism. A rare but deadly poisoning from bacteria that grow without air in low-acid foods. Prevented by pressure canning and proper acidity.

Bloom. The natural coating on a fresh egg's shell that keeps germs and air out.

Brine. Salty water. In fermenting, the vegetables' own juice mixed with salt.

Burr. A tiny curl of metal on a knife edge that shows sharpening has reached the edge.

Cold smoking. Smoking below about 90 F to flavor and dry food without cooking it.

Condensation. Water vapor turning back into liquid when it touches a cool surface.

Cracklings. The crisp bits left after rendering fat.

Deep litter. Letting coop bedding build up and compost in place over winter.

Faraday cage. A closed metal box that blocks electromagnetic waves.

Fatwood. Resin-rich pine wood that lights easily, even when damp.

Ferro rod. A ferrocerium rod that throws very hot sparks when scraped.

Graft union. The point where a fruit tree's top was joined to its roots.

Headspace. The empty space between food and lid in a canning jar.

Heirloom. An open-pollinated plant variety whose seeds grow true to the parent.

Kahm yeast. A harmless white film that can form on ferments.

Leaf lard. The finest pork fat, from around the kidneys.

Lye. A caustic alkali, once made from wood ash, used to make soap.

MPPT. A type of solar charge controller that gets the most power from a panel.

Pickling lime. Calcium hydroxide, used in water-glassing eggs and some pickles.

Poultice. Soft, moist plant matter applied to skin.

SODIS. Solar water disinfection: clear bottles left in strong sun.

Tallow. Rendered beef or mutton fat.

Tinder. Fine, dry material that catches a spark or flame first.

Water glassing. Storing eggs in a lime solution to keep them for months.

H. WHERE TO LEARN MORE

- **Food preservation:** the USDA Complete Guide to Home Canning and the National Center for Home Food Preservation (nchfp.uga.edu).
- **Emergency preparedness:** Ready.gov and your county emergency management office.
- **First aid:** American Red Cross first aid/CPR classes and the free Stop the Bleed program (stopthebleed.org).
- **Gardening:** your state's university cooperative extension office, which offers soil tests and local planting calendars.
- **Radio:** the ARRL (arrl.org) for amateur radio licensing and local clubs; the FCC for GMRS licenses.
- **Foraging:** a regional field guide and a local plant walk led by an experienced forager or naturalist.

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MY SKILLS LOG

Tick off each skill once you've actually done it, start to finish.

#	SKILL	DATE TRIED	DONE
01	Storing a Year's Supply of Water		☐
02	Catching Rainwater from Your Roof		☐
03	Making Water Safe: Boil, Bleach, or Sun		☐
04	The Bucket Sand Filter		☐
05	Fresh Water from Seawater		☐
06	Gathering Water from Air and Dew		☐
07	Pemmican: The Original Survival Food		☐
08	Keeping Eggs a Year Without a Fridge		☐
09	Canning Meat in a Jar		☐
10	Rendering Lard and Tallow		☐
11	The Backyard Smokehouse		☐
12	Sun and Air Drying Fruits and Vegetables		☐
13	Sauerkraut and Crock Fermenting		☐
14	The Clay Pot Refrigerator		☐
15	Yeast-Free Soda Bread		☐
16	Starting and Keeping Sourdough		☐
17	Hardtack: Bread That Lasts Decades		☐
18	Depression-Era Meals That Stretch a Dollar		☐
19	Cast-Iron Care (and the Mistake Everyone Makes)		☐
20	Churning Butter in a Jar		☐
21	Twelve Uses for Eggshells		☐
22	Stopping Serious Bleeding		☐
23	Building a Home Medical Kit		☐
24	Oral Rehydration Solution		☐
25	Fire Cider and Honey Garlic		☐
26	The Plantain Poultice		☐
27	Elderberry Syrup		☐
28	Cooling a House Without Electricity		☐
29	Staying Warm in a Winter Blackout		☐
30	The Rocket Stove		☐
31	Light Without Power		☐
32	Ten Uses for Wood Ash		☐
33	A Trash-Can Faraday Cage		☐
34	Simple Solar Backup Power		☐
35	Feeding a Family on a Quarter Acre		☐
36	Building a Pallet Garden Fence		☐
37	Dwarf Fruit Trees in Small Spaces		☐
38	Composting the Easy Way		☐

#	SKILL	DATE TRIED	DONE
39	Saving Your Own Seeds		☐
40	Keeping a Few Backyard Hens		☐
41	Building and Lighting a Campfire		☐
42	Fire with a Ferro Rod		☐
43	The Dakota Fire Hole		☐
44	The Debris Hut		☐
45	Five Knots Worth Knowing		☐
46	Finding Direction Without a Compass		☐
47	Signaling for Rescue		☐
48	Escaping a Sinking or Trapped Car		☐
49	Spotting Hidden Cameras in a Rental		☐
50	Situational Awareness		☐
51	Handling Insults and Difficult People		☐
52	Building Calm Confidence		☐
53	Your Family Emergency Plan		☐
54	Radio When the Phones Go Down		☐
55	Fixing a Stuck or Broken Zipper		☐
56	Sharpening a Knife on a Stone		☐
57	Sewing a Button and Mending a Tear		☐
58	Fixing a Dripping Faucet		☐
59	The Essential Home Tool Kit		☐
60	Darning Socks and Patching Jeans		☐

NOTES

Your own recipes, measurements, garden records and lessons learned.

THE BOOK YOU NEED TO TAKE BACK YOUR INDEPENDENCE

A century ago nearly every household could keep its water clean, put up a winter's worth of food, nurse a fever, heat a home without the power company and fix whatever broke. Most of that knowledge quietly slipped away in just two generations.

The Independent Household brings it back: sixty clear skills, each laid out on a single page with step-by-step pictures, a materials panel and notes on what can go wrong. Tabs on every page help you find what you need, even by lantern light.

INSIDE YOU WILL LEARN HOW TO

- Store a year of water and purify any source
- Can meat, water-glass eggs and make pemmican
- Bake soda bread, hardtack and sourdough
- Stop bleeding and build a home medical kit
- Cook on a rocket stove and beat a blackout
- Build a Faraday can and a solar backup
- Grow food on a quarter acre
- Make fire, shelter and signal for rescue
- Escape a sinking car and spot hidden cameras
- Sharpen, sew, darn and fix a dripping faucet



"Use it up, wear it out, make it do, or do without."

OLD AMERICAN SAYING

THE INDEPENDENT HOUSEHOLD

Self-Reliance / Homesteading / Reference

HOME EDITION · PRINT AT HOME